Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 — Energy storage devices, hydrogen energy, and sensors · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

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· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_03 · Energy storage devices, hydrogen energy, and sensors

1 UE26CY151A · Unit_03: Energy storage devices, hydrogen energy, and sensors

This unit master organizes source material for Engineering Chemistry, Unit_03, officially titled Energy storage devices, hydrogen energy, and sensors. It includes 10 video transcripts and 12 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26CY151A · Unit_03: Energy storage devices, hydrogen energy, and sensors

Official catalogue entry: UE26CY151A, Unit_03 (Energy storage devices, hydrogen energy, and sensors). Bucket folder present: yes. Videos: 10; slide decks: 12.

EV0000001

2 Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I)

Study focus: Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll start with unit 3, so that is module 5, and we are going to learn about energy storage devices.

Key details

- Then we learn about modern batteries like zinc air battery, lithium batteries, and specifically lithium ion battery.
- Then we study about reserve batteries and then fuel cells.
- principle working and efficiency of fuel cells.
- We also look at three different types of fuel cells.
- Hydrogen oxygen alkaline fuel cells carbon monoxide oxygen solid oxide fuel cell and methanol oxygen polynomial membrane fuel cell.
- Then we talk about oxygen sensors.
- And then supercapacitors and finally Ragoni plot.
- Battery is a device which stores chemical energy and converts it into electrical energy on demand.
- We said that they are devices which convert chemical energy stored in the reaction into electrical energy.
- Another property of battery is that they come in variables, shapes, and sizes.
- You can have batteries which are very tiny like the ones which are used in hearing aids, or you can have batteries which are very huge like.
- They could weigh up to 200 tons and they are used in submarines.
- And also the shapes of the batteries can be varied depending on the application.
- And the third most important point about batteries is that they are portable.
- You can carry them into space, you can use them on the land, and you can also take them underwater.
- Three important properties of batteries which make them very useful.
- We all know that there are a lot of applications of for batteries like in consumer goods.
- We have mobiles, laptops, calculators, flashlights on one hand.
- And on the other hand, we have very specific applications like in emergency lighting.
- space vehicles, electric vehicles, and pacemakers.
- We know that a battery is a galvanic cell, therefore we will require the two electrodes, anode and cathode.
- What material would I choose for the anode?
- It has to be something which undergoes oxidation very easily.
- Therefore, we find most of the batteries they use metals as the electroactive material at anode.
- And most commonly used materials for batteries are zinc, lead and lithium.
- Most of the batteries now are being made with lithium.
- And we also need a cathode where reduction takes place.
- We know that oxidation and reduction should happen simultaneously.
- And please remember this that a metal can never undergo reduction.
- It can undergo oxidation.

Example or application from the lecture

- Let us look at the various components of a battery.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

But it can never undergo reduction for the simple reason that it's already in the lowest reduced form.

Key details

- Metal in the form of iron.
- That means in the form of compounds.
- Metal itself is used as the anode, but at the cathode we use compounds of metals which can undergo reduction easily.
- Then MnO2 is also an oxide which is used.
- Not only oxides, we can use chlorides, nitrates, any of the compounds, but most commonly used are oxides.
- And we can even use oxygen as the cathode active material because we know that oxy can undergo reduction easily.
- What would serve as an electrolyte?
- Substances which have very good ionic conductivity.
- Ionic conductivity means we need ions which will move very easily from one electrode to the other.
- We can have acids, we can have alkali, we can have salt solutions, we can also have molten salts.
- And also because molten salt can produce ions which are free to move.
- And we can also have solids which are doped oxides.
- And then we can have polymers also which can be used as electrolytes.
- NAFion is a polymer membrane which is used in some batteries.
- Then we come to the last component that is a separator.
- What is a separator?
- It's usually an insulator which is going to separate the anode and the cathode compartment.
- It is to prevent the anode and cathode coming in contact with each other because if that happens, there's going to be internal short circuit.
- To separate the anode and the cathode.
- But at the same time, it should be able to transport the ions because apart from the oxidation and reduction which is happening at the anode and cathode, respectively, this transport of ions in the through the electrolyte is very important for the battery to function.
- Therefore the electro the separator should be able to transport the ions from anode to cathode and cathode to anode.
- There are many types of separators which are employed based on the design of the battery.
- But this anode active material may be in the form of a powder or a paste and therefore we need what we call as a current collector.
- Always there will be a current collector.
- The current collector itself is not involved in the reaction.
- It is the it is the electroactive species which is involved in the reaction.

- And then on the cathode health also we have a similar.
- Electroactive species, but this is the cathode electroactive species, and this will be the current collector.
- active material and here we have the anode active material.
- Then what do we need next?
- We use the anode material, we use the cathode material, and then we need the electrolyte.
- It is present between the anode and cathode and it is going to make sure the anode and cathode do not come in contact.

Example or application from the lecture

- And most of the times we find lead oxide is a good example, like in your lead acid battery.
- So for example, we can have H2SO4, we can use H2SO4 or KOH or Nafion.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

Okay and then we also have terminals here, which you are aware because we all use a lot of batteries.

Key details

- we have placed inside the battery.
- So inner battery, therefore, so both are terminals here.
- And this whole thing is now kept inside a housing it is called it's a container into which this anode and cathode reactive materials along with the current collectors are placed and this is filled with electrolyte and there's a separator separating the anode and the cathode.
- Now the battery can discharge because we know it is going it is a source of energy, it is source of power.
- What do you mean by discharge?
- It is delivering power.
- When the battery discharges, we it is delivering power.
- So therefore it acts like a galvanic cell.
- Again, galvanic cell uses the chemical energy and converts it into electrical energy.
- Therefore, when the battery is discharging and giving us power, the battery.
- Bacteria is behaving like a galvanic cell.
- They move through the external wire and at the cathode we have some species say A, which is going to take up these n electrons, and we get A N minus.
- We have the elect the electrons being generated at the anode.
- They move through the external wire thereby generating power.
- The power there's a flow of current and whenever electrons flow, then we say it is it is going to it is giving you current.
- The power will be provided to that load.
- And then here the at the cathode the electrons are being taken in on absorbed and that gives rise to an minus.

- They are going to move in the opposite direction.
- Again, you have to remember electrons never come through the solution.
- They go through the current collectors, pick up these electrons and through the external wire we see that the electron movement will happen like this.
- Only certain batteries we can charge them back to the original condition.
- While charging the battery is consuming power.
- The battery consumes power and now it is going to behave like an electrolytic cell.
- When we provide power and the.
- Reaction takes place, a non-spontaneous reaction takes place.
- So this is going to act as electrolytic cell because we are providing power.
- You can see there's a battery which is connected here, and now what happens at the anode.
- oxidation has to take place.
- Now what happened is your MA because of your you're giving this power over here the MA now undergoes electrolysis and it breaks into Mn plus and An minus.

Example or application from the lecture

- Let us see.

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EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

The anion AN minus is moving towards the anode and at the anode it forms a and n electrons are given out.

Key details

- And at the cathode what happens?
- The cations move towards the cathode and here you have they take up the electrons.
- The cations will take up the electrons.
- And why is this negatively charged?
- Because you have connected it to the negative terminal of the battery, and this is positive because you connected it to the positive terminal of the battery.
- It is going to take up electrons.
- And then it is going to form M.
- Because of the charging process.
- MA is again.
- become M plus A.
- How did what happened what happened to this MA actually, how did we get this MA?
- Because MA underwent electrolysis produced.
- A n minus and m n plus which move towards the respective electrodes, and that is how we got back M and A.
- And therefore, M A is giving you M plus A is the overall reaction.
- The battery now has because of the charging, we have been able to regenerate the electroactive species, so it is ready for the next cycle where it can discharge and deliver

power.
- When it is discharging, it is behaving like a galvanic cell.
- When it is charging, it behaves like a electrolytic cell.
- One is this primary battery.
- Once the electroactive species are consumed, the battery has to be discarded.
- So here the battery can act only as a galvanic cell, that means we cannot charge the battery to get back the electroactive species.
- And examples for primary cell would be dry cell and lithium MNO2.
- Discharge the battery, use the power, then again charge it back so that you get the electroactive species.
- So therefore we can say it behaves as a galvanic cell as well as electrolytic cell.
- When it is discharging, it is a galvanic cell, when it is charging, it is an electro.
- Electrolytic cell, and they are also called as storage batteries.
- Secondary batteries are also called storage batteries because we are regenerating the electroactive species and storing them inside, and then whenever.
- There is demand the battery is discharging and giving us power.

Example or application from the lecture

- Example would be the most famous lithium ion battery, then we have the lead acid battery, and then we have those nickel cadmium batteries which were called as the nicad batteries.
- Suppose we want to use a battery for a particular application, there are certain characteristics we will look for and that we are going to study under the heading of battery characteristics.
- Now the first and foremost let us study about the voltage of a.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

Which is very important.

Key details

- When we want to take a battery or select a battery for a particular application, we have to look for what is the voltage we require for that particular application.
- We know that voltage is basically the EMF generated from the cell.
- Free energy of the.
- For each reaction, there's a different free energy delta G.
- The first and foremost we see is E cell, so E naught cell.
- The difference between the cathode electrode potential and the anode electrode potential.
- Remember we are always talking of standard electrode reduction potentials.
- And what the difference between the two will decide what is E0 cell of the battery.
- And higher the difference, more is going to be E0 cell, and therefore in return E cell will be a very high value.
- How will we make a make sure E naught cell is high?

- By choosing the right electrodes.
- We can have a high E naught cell value.
- Now coming to the next that is temperature.
- You can see from here that the voltage of the cell is going to decrease.
- What is q basically reaction quotient means concentration of the products divided by the concentration of the reaction.
- Remember there's a reaction which is happening inside the battery, so the concentration of the products are continuously.
- Increasing and that of the reactant is decreasing when the battery when the forward reaction is happening and the battery is generating power.
- So this value Q keeps changing continuously, but the voltage will not change to that extent because then it is in a log term.
- So because log q is involved in this term.
- It's mostly decided the voltage of the battery is mostly decided.
- By E0 cell.
- Nevertheless, as the battery reaction continues, we will find that there is a change in the E cell value, but it's only marginal.
- That is about voltage.
- Battery characteristic again very important and that is current.
- It's a measure of the rate at which the battery is discharging.
- And what is that something here?
- The rate at which the battery is discharging means what?
- The rate at which the current is being generated.
- The rate at which the current is flowing.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 25:00–29:59

If I want to get a good current from that battery, we cannot change the voltage because voltage depends on the choice of electrodes.

Key details

- I should keep the resistance to minimum.
- Fast and another way of keeping the resistance low is keeping the inter electrode distance less, meaning that the anode and cathode have to be placed.
- close to each other so that the inter electrode distance is less and also the electrolyte should have good conductivity so that the ions will move easily from one electrode to the other.
- Apart from that.
- One more important thing which should happen is that so all this we are doing basically for a rapid electron transfer reaction because if the electrons are generated fast, then the current will be more.
- If you there's a very slow generation of electrons then obviously the current flow is will not be that high.

- from kinetic study that more is the concentration of the reactant, more is going to be higher is going to be the rate of the reaction.
- The same thing applies here.
- If we have a large quantity of electroactive species, the reactants are more in the quantity and therefore the electron it can undergo oxidation or reduction faster the rate.
- will be fast here and therefore the electrons will be generated fast.
- So we say rapid electron transfer is going to happen and therefore we can get a good flow of electrons.
- Therefore the current which is the rate at which the electrons are flowing is.
- Going to be very high.
- In the next class, we are going to look at many more interesting battery characteristics.

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EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

3 Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II))

Study focus: Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000026 · EV0000027

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module five on energy storage devices.

Key details

- We are talking about batteries.
- We discussed two battery characteristics that is voltage and current.
- And some of them which we are going to study today are capacity, energy storage density, cycle life, energy efficiency, shelf life, tolerance to service conditions, power density and energy density.
- What is the definition?
- The charge or amount of electricity that can be obtained from the battery is known as capacity of the battery, and the unit in which it is expressed is ampere hours.
- W is the mass of the electroactive material.
- M is molar mass of the electroactive material.
- And n is the number of electrons being exchanged in the reaction.
- 2 plus oxidation should happen at dianode.

- N is going to be 2, and F is the Faraday that is 96500 coulombs per mole.
- We can use to determine the capacity of a battery if we know the amount of the electroactive material present at the electrode.
- How can we say that?
- Because W is figuring in this expression.
- And what is W?
- Just now I told you W is the amount of the electroactive material.
- Bigger is going to be the battery.
- Therefore, we can say that the size of the battery determines the capacity of the battery.
- Now, another condition on which the capacity of the battery depends is the discharge of the battery.
- What does it mean?
- How much current is being drawn from the battery.
- It is not very different from what we have studied before.
- Because capacity is a charge contained in the battery.
- And current multiplied by time is also charge.
- Wnf by M.
- If we know how long the battery lasts when we are drawing a certain amount of current, then we can use this expression to determine capacity of the battery.
- We are seeing how the voltage varies with time for a particular battery.

Example or application from the lecture

- Let us first learn about capacity of a battery.
- The capacity is also given by a current into time.

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EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 05:00–09:59

almost a flat portion and after that it comes to a pos part the voltage is now too small for the battery to.

Key details

- Perform the required function.
- Initially, there will be a drop in the voltage, and then we see that it gives a steady voltage for a long time until it reaches a point.
- After which we declare that battery as dead, if it's a rechargeable battery, we need to recharge it.
- So this is the cutoff voltage because beyond this, every device requires a certain voltage.
- Device is not going to function using this battery, so either it needs to be replaced or it needs to be recharged.
- So we can say that the capacity of the battery, because what we are looking at when we are drawing a certain amount of current, how long the battery will last.
- portion of this curve can give us an indication as to what is the capacity of the battery.

- So if this flat portion is very long, we say that the battery has a good capacity because at this particular current it is giving you it lasts for a longer time.
- If this portion is the flat portion is less then we say that the capacity is not so good.
- Energy storage density is defined as the amount of charge or electricity per unit weight which the battery can hold.
- It is telling the amount of charge.
- Which the battery can hold, but it is now it is become density.
- We are calling it density, so that is per unit weight which the battery can hold.
- So that means if we want to get a good electricity storage density, we have to use the right material for all the components.
- We say that the designing of the battery is very good.
- Electroactive material that is taking part in the reaction at the cathode or the anode.
- But here when we say weight, we refer to weight of the battery on the whole, meaning all the components of the battery have to be taken into account.
- Why lithium is such an important anodic material?
- Why is it such a popular anodic material?
- Because we know it's a lightweight metal.
- Just seven grams of lithium can give you one faraday of charge.
- Whereas if we use lead, we know that lead is also a very popular anonic material.
- like lead acid batteries etc which are which are used quite a lot so but here we see that we require 104 grams of lead to produce the same amount of charge that is one farity so therefore lithium has become a very popular aerodic material.
- Now coming to the next battery characteristic, cycle life.
- Such batteries are referred to as secondary batteries.
- These secondary batteries can be again and again charged, but they also have a lifetime.
- The electroactive when it is when the battery is discharging, the electroactive material is consumed.
- But when the electroactive material is completely consumed, we can again.

Example or application from the lecture

- Now, let us come to the next battery characteristic that is energy storage density.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 10:00–14:59

recharge the battery and when we recharge the battery this electroactive material is getting regenerated.

Key details

- That means the cycle can go on and on because you can charge and discharge, charge and discharge as many times as we want, but we know that this is not true.
- Every secondary battery also has a life.
- How will we define it?

- The number of charge-dischar cycles that are possible before the battery fails.
- How many times can we charge and discharge the battery?
- That 500 times the battery can be charged and discharged.
- The battery can go through before it fails.
- Why do secondary batteries have a limited cycle life?
- Theoretically, you can charge and discharge, charge and discharge, but this does not go on forever.
- We know that there is a limited life for a particular battery, even though it is a secondary battery.
- For this is corrosion of the contact points.
- We know that we are using as electrolytes in the form of acids, alkali, salt solution, all those are very corrosive materials, and therefore the contains with time could get corroded.
- Reason why we see that the batteries have a limited life.
- Going to come out and electrons will be generated and the electroactive species is getting consumed.
- has to come back and form a nice layer around the around the current collector.
- Initially, yes, there will be nice layers of this electroactive material, and you will get something like in this form.
- But as time progresses, you go on using the battery for a long time.
- We find that this material instead of being on the current collector it will start to shut down.
- This electroactive material during the charging process has not gone back to the current collector and therefore what will happen the this material which has shut down is not available for the next discharging cycle.
- The electroactive material is.
- Able to undergo reaction as long as it is around the current collector.
- It is available for the discharging.
- But once it is not on the current collector and a shutdown like this, the electroactive material slowly goes on decreasing, and we find that the battery will become weaker and weaker.
- So first we talked about corrosion, next.
- What do we mean by this?

Example or application from the lecture

- Let us draw a battery here and see what exactly happens inside the battery.
- Suppose the battery is discharging, all this is the electroactive species present around it, and this is the current collector.

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EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 15:00–19:59

When the battery is discharging, we find this electroactive material gets away from the current collector.

Key details

- When you are charging, again as I said in the previous point, this material has to come back to the current collector.
- Forming nice layers around the current connector, we find that with time these dendrite like growths start.
- the battery shows tree like growth or they are called dendrite like growth.
- Anode, the same thing can happen at the cathode also.
- Even there is a separator in between, but they may pierce through the separator and what we find is what will happen when cathode and another come in contact with each other, there is going to be short circuiting and the battery is going to fail.
- The electroactive material did not get back onto the current collector.
- Instead, we find dendrite like growth.
- Or you can simply say tree like structures.
- Have to maintain low inter electrode distance.
- Coming to the next battery characteristic, energy efficiency of a battery.
- Percentage of energy efficiency would be energy released on discharge by the energy required for charging multiplied by 100.
- Charging this is again applicable to secondary batteries.
- Or this definition of energy efficiency is applicable to secondary battery.
- It is never 100%.
- And this efficiency depends on the efficiency of the electrode reactions.
- If there is any hampering in the electrode reactions, or they are going slow, or there is some kind.
- Problem the anode efficiency, the cathode efficiency is not good, then we find that the energy efficiency also falls down.
- Coming to the next characteristic that is shelf life.
- Many times we keep batteries at home, and especially the dry cell, we find that when we want to use the battery, it is not working anymore.
- That what is it, what is responsible for the battery being dead when it is not at all being when it has not even been used.
- That characteristic is called shelf life.
- Shelf life, how will we define shelf life?
- The maximum time, the maximum time for which you can store the battery, and yet it will give you the specific performance.
- How long can you keep the battery without using it?
- And yet it gives you the specific performance which is required.
- But many times we find that batteries are dead by the time we want to use, and of course, this is we are talking of a primary battery like dry cell, we find that.
- They are not useful anymore and they cannot be used anymore and have to be discarded.
- What could be the reason for this?
- The reason is there is something called as self-discharge, which happens inside the battery.
- The moment the components are assembled.
- the reaction in the battery starts.
- So therefore, or there could be other secondary reactions also happening which are going to consume the electroactive material and therefore when we leave it for a long time without

using it then the battery all the material could have been used up or there could also be corrosion and leaking of the battery.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

Core explanation · 20:00–24:59

the shelf for a battery having shelf life.

Key details

- Of course it is very important for a primary battery to have a good shelf life.
- Why because so that even if it is stored for some time we can still use it as primary batteries are one time use batteries.
- For secondary batteries also have a shelf life, but the good thing about secondary batteries is they can be recharged.
- Problems like corrosion and all will of destroy the battery.
- Now coming to the next point, tolerance to service conditions.
- The battery has to be tolerant to different service conditions.
- Be available, I mean it should operate at very low temperatures or at the high temperature at which the.
- The motor gets heated up, I mean the motor is going to heat up the machine, and therefore it has to tolerate that high temperature.
- or shock because when the when you're moving the car when the car is moving on the road or in any terrain there will be lot of vibration and shocks which the battery will also have to face.
- So therefore the battery should be designed such a way that it should be tolerant.
- To whatever service conditions you are going to use it for, whatever application it is going to be used for, the service conditions, the battery should be able to tolerate those service conditions.
- Now energy density is defined as the amount of energy which is available from the battery divided by mass of the battery, or you can say weight of the battery.
- Again, when we.
- We say density, we should remember the weight includes weight of all the components of the battery.
- Divided by weight of the battery.
- And the what is the current the battery is giving us multiplied by the voltage at which it is operating, multiplied by the time for which the battery will.
- Watt hours per kg, or we can say kilowatt hours per kg.
- Depending on what is the application, the unit can be varied, but nevertheless, it is power into time divided by weight of the battery.
- Coming to the last battery characteristic that is power density.
- Pow density is defined as the power available from the battery divided by mass of the battery or weight of the battery.
- Again since we are talking of density.

- It should be the weight of all the components of the battery or mass of all the components of the battery.
- Power is always current into voltage.
- being you have used there multiplied by voltage of.
- The battery divided by weight of the battery, mass of the battery.
- That would be power density of the battery and the unit will be watts per kg.
- I said for energy density, it can also be kilowatt per kg, watts per gram, depending on the application and depending on the size of the battery, and depending on what power it is generating.
- We can choose the correct unit, but the unit is going to be that of power divided by.

Example or application from the lecture

- For example, if there is a variation in temperature.
- The temperature range has to be very large for, for example, the car batteries, and apart from that, they should also be tolerant to say vibration.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 25:00–29:59

weight of the battery.

Key details

- Both look very similar.
- But power density, as we know, energy is the work done.
- Energy is always work done, and power is at what rate the energy is being produced.
- When the car is going uphill, you need a high power density.
- And when would you require a good energy density?
- How would whether the energy density is good?
- it lasts for a long time before it requires charging.
- Power density will be when it the when the say it is in a car and how long so energy density is how long the car would go before it requires charging and power density.
- How steep a hill can it climb?
- That would be determined by its power density.
- Nevertheless, we have to know.
- See what is the application.
- Some applications could require a very high power density, but some applications could require a low power density.
- Also, energy density or power density, the requirement could be that we need a certain amount of energy density for a long period of time or.
- a high energy density for a very short period of time.
- Whereas if you're running the car, if you're running an electric vehicle, then you would require a high energy.
- Density, a constant energy density for a long time.

- density and all these battery characteristics are very important and it's not necessary that all these characteristics should be at the high value for every for an application.
- Sometimes we require.
- low power density is fine.
- But then there are other requirements like it should be able to tolerate the body temperature, then it should have it should be very accurate voltage.
- And it should be very, very reliable.
- It should last for a long time.
- So in the next class.
- We are going to look at some modern batteries and then we will see how all these battery characteristics have been used for these.
- or what are these battery characteristics or that particular battery.
- So in the next class we will talk about zinc air battery.

Example or application from the lecture

- When would you require a high power density when you when you say for example when you're when the car is going uphill.
- For example, in the flash of a camera, we need very high energy density, but for a very short period of time.
- For example, in a pacemaker, we require we don't require very high power, we require just few micro watts.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

4 Numericals_Energy storage devices

Study focus: Numericals_Energy storage devices. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000050 · EV0000051

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module five on energy storage devices and we'll try to solve some numericals today on this topic.

Key details

- After that, you also need to find out what is the weight of the entire battery.
- So here first we have to find out the weight of zinc.
- And then we have to find out the weight of the entire battery.
- I into T, that is current into time.
- And two hours is the time.

- It is working for it is lasting for two hours.
- That is we are converting it into seconds.
- We need to be very careful with the units.
- In this case it is zinc.
- That is the anodic material.
- N is the number of electrons getting exchanged.
- We know from the reaction it is two electrons.
- So therefore C is equal to Wnf by M.
- From here we can find out the weight of the electroarms.
- material by rearranging this equation.
- That is the reason why we found out the capacity.
- In this case, we could determine capacity from here, and this formula is used to determine weight of the electroactive material.
- So, therefore, capacity is known.
- into 652 multiplied by divided by two multiplied by nine six five zero and we have found out the weight of zinc it is three point zero three grams of zinc.
- Now coming to the next part where we want to find out the weight of the entire battery.
- Amount of charge that can be stored in the battery per unit weight of the battery.
- And then there we said weight of the battery, when we say per unit weight of the battery, that is weight of the entire battery, including all the components.
- That is electricity storage density because we said the amount of charge that can be stored in the battery.
- And what is the amount of charge that can be stored in the battery?
- That is equal to the capacity of the battery.
- Only thing you need to be very, very careful with the units.
- The unit is where you have to be very careful.
- You have to give your answer in ampere hour per kg.
- This is the requirement of this numerical atomic mass of lithium is seven.

Example or application from the lecture

- Let us take up the first numerical on battery characteristics.
- And for this electricity storage density is given to us and molar mass of zinc is known, 65 grams, and F, that is Faraday's constant, is to be taken as 96500 Coulombs per mole.
- Let us see the solution.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 05:00–09:59

Capacity is either current into time or it can be determined by WNF by M.

Key details

- So because no current or time is mentioned here, we have to go with the other formula that is C is equal to Wnf by the weight of the battery.

- It is 2 grams.
- Lithium is always we know lithium is monovalent so plus one electron therefore here we have one electron that is being exchanged in this and 9650 divided by seven that is the atomic mass of lithium.
- But this is in Coulombs.
- Remember capacity is always what when we calculate it will be in Coulombs and Coulombs is always ampere second.
- So therefore if you want to convert it into ampere hours what do you need to do?
- Divide it by three six zero and that would give you an answer seven point six five eight ampere hours is the capacity.
- Weight of the battery.
- Please read carefully.
- This is the weight of the electroactive material which is used to calculate capacity of the battery, and this weight is weight of the entire battery.
- It is the total weight of the battery and it contains 5 kg of lead as the anode material.
- for 10 hours.
- The time is known to us.
- Number of electron transferred in the redox reaction is 2.
- These are simple.
- elements for which may you should know what is the simple reaction when it undergoes oxidation.
- Since it's the unknown material, we write the reaction this way.
- We need the energy.
- Energy density and we need the power density.
- What do we have?
- So that means I can straight away determine the capacity here using C is equal to Wnf by M.
- Yes, so we will use C is equal to Wnf by M that is 5 into 10 raised to the power of 3.
- Why did we do this?
- Because we are now expressing in grams.
- 2 is the number of electrons that is getting exchanged over here and 9650 is zero is known to us.
- What is the value in ampere second?
- This value which you got from here is ampere seconds.

Example or application from the lecture

- Let us look at the solution.
- So we have to use this formula for capacity, WNF by M, because the weight is of lithium, which is used as the anodic material, is given to us.
- Let us see another type of question where we need to find out the energy density and the power density.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 10:00–14:59

60 into 60.

Key details

- Because straight away you can't write this and divide by 3600.
- It doesn't make sense.
- if you write this as ampere seconds and then you show it as ampere hours that would make more sense.
- Doesn't matter, we can always write it like this.
- divided by weight of the battery.
- If it was asked in a grams, then you to need to convert this into grams.
- But we neither we don't we know the time but we don't know the current.
- But is there any other way I can use this formula?
- Yes, we know that I into t is the capacity of the battery.
- And that is exactly what we calculated here.
- That is employed, and then we divide it by 20 weight of the battery, and we get 129.4 watt hours per kg is the energy density.
- Understand, we don't know the value of current, you need not know also, because once the capacity.
- You can straight away in this formula use capacity.
- will be energy density divided by time because it is current into voltage divided by.
- Current into voltage divided by.
- The weight of the battery.
- That is the formula for power density.
- Current into voltage divided by weight of the battery.
- Current into time into voltage by weight of the battery.
- Power density, all you have to do is if the energy density divide it by time and you get your formula I into v by w that's exactly what we have here.
- The time is known to us as 10 hours.
- This answer is also in hours.
- Instead we wanted it in hours because the time is in hours, they get the units get hour gets cancelled.
- here and therefore we'll end up with 129.4 divided by 10 and the answer would be this is also hours this is also what hour per kg so therefore you end up with watts per kg 12.9 watts per kg is your answer for power density.
- The efficiency of a fuel cell.
- We know that in an alkaline fuel cell.

Example or application from the lecture

- Volta is given to us as two volts here.
- And what is given to you E naught cell is given to you and delta H formation of water is equal to minus two eighty five point eight kilojoules per mole.
- Energy being given out.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 15:00–19:59

with hydrogen and oxygen.

Key details

- H2O since we need only one mole, so this is how we write, and we know if you we know that if we write it as anode and cathode reaction, the number of electrons is going to be two electrons which will be exchanged in this reaction.
- Efficiency percentage is equal to delta G by delta H multiplied by 100.
- So when you substitute here and because we have to find out the percentage multiplied by 100.
- Minus N F E.
- We have to calculate the heat evolved during the reaction.
- Efficiency is again in the form of percentage.
- Efficiency percentage is delta G by delta H multiplied by 100.
- N again we know is going to be.
- Two for this particular reaction, what is the reaction happening here?
- Alkaline fuel cell H two plus half O2 is giving you.
- The delta H value is to be found out multiplied by 100.
- That's you will get it in joules, but then you want to determine in kilojoules.
- Both the reactions.
- When you say delta H combustion of hydrogen, when you say Delta H combustion.
- Or you can say delta H formation of one mole of water.
- In that case also the reaction remains the same, therefore the delta H will be same.
- So whether you say delta H formation of water or you say delta H combustion of hydrogen the value will remain the same and both of them here we see it is one mole therefore delta H is equal to minus 286 kilojoules per mole.

Example or application from the lecture

- Let us look at the solution.
- Now since here E naught cell is given to you and not delta G, we need to first calculate delta G.
- So this would give us the value of delta G and delta H is given to us minus 285.8 and because it's in kilojoules we will convert it into joules.

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EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 20:00–24:59

You need to know the efficiency and you also need to know what is the E naught cell, or directly if the delta G also, you will be able to use this formula to calculate.

Key details

- So in the next class, we are going to start with the next unit that is about corrosion chemistry.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

5 Zinc -air battery

Study focus: Zinc -air battery. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000070 · EV0000071

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module 5 on energy storage devices.

Key details

- That is, we are talking about batteries.
- For the Z care battery we look at the construction, working, advantages, disadvantages, and applications.
- So zinc air battery can be classified as a metal air battery because as you can see in the name the zinc and the air, so it is metal air batteries.
- Another important fact about this battery is that it's an alkaline battery.
- The most important thing about this battery is that it is using oxygen directly from the atmosphere to produce electrochemical energy.
- Property of this battery is that the energy density is going to be very high.
- The energy density for such batteries is very high.
- Battery has been considerably reduced.
- The reason being that one of the electroactive material that is the cathode electroactive material which is oxygen need not be stored inside the battery and can be taken directly from air.
- Salient feature in fact of this battery is that it has high energy density.
- For in a battery, in the first class when we talked about batteries, I told you there are four components, basic components of a battery, the anode,.
- The cathode, the electrolyte, and the separator.
- We have already studied just now that zinc granule zinc is the anode active material.
- In what form is it present?
- Zinc is taken in the form of granules, and why would that be to increase the surface area so that the reaction can happen at a faster rate.
- The role of this gelling agent is to hold the granules together.

- It is activity material is oxygen, but then we know that there needs to be a current collector.
- Just having the electroactive material is not sufficient.
- Yes, oxygen will be taken from the air, but we need to have electroactive material.
- electroactive material is taken from the air, but we need to have a current collector.
- Responsible for picking up the electrons or providing the electrons for reduction in this case because it is the cathode.
- We have carbon.
- That is in the form of crepite.
- That is because.
- At the cathode, oxygen has to undergo reduction, which is a slow process.
- In order to expedite the process, in order to make the reaction happen faster, we need to use a catalyst.

Example or application from the lecture

- Let us look at the construction of such a battery.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 05:00–09:59

Why would that be required?

Key details

- We are taking oxygen from the atmosphere.
- So therefore, in order to.
- Remove the moisture, not allow the moisture into the battery, we need to add a wetproofing agent, and all this is now coated on a nickel wire mesh support which is acting as the current collector.
- And apart from this, we also have an outer.
- layer of air permeable teflon layer.
- Now what would this be what would be the role of this teflon layer it's very obvious that teflon layer will pr prevent the moisture from entering into the battery, but at the same time it has to be air permeable also because it has to allow the oxygen to enter the battery.
- Though there is no cathode active material still there has to be a cathode inside the battery.
- Of course this cathode is going to form a very thin layer in the a thin small part of the battery and the remaining part will be the anode.
- through this air permeable Teflon layer, so we have air access holes on the cathode side, which are going to provide the pathway for oxygen to enter the battery.
- Why it has to be like this.
- The separator is soaked in electrolyte.
- We'll learn about it in a few minutes now.
- That is the construction.
- Positive that is the cathode.
- Through these holes.

- So this has to be cathode because oxygen undergoes reduction.
- the hydrophobic membrane that is your teflon layer, which we mentioned just now.
- Carbon and catalyst also to in order to for one carbon is for conductivity and catalyst is to speed up the reaction.
- Soaked in electrolyte.
- Separator soaked in electrolyte.
- So we can see that we because oxygen is being taken from the.
- Atmosphere directly, so it need not be stored inside the battery.
- Therefore, the entire battery space is actually filled with the anode active material and that is zinc.
- And then we have a cap here the metal cover which is the act acting as the negative terminal and we have a cup over here which is going to act as the positive terminal.
- The gasket is going to.
- Prevent the anode and cathode coming in contact with each other.
- At the same time, it is going to seal the two metal plates together, the metal cup and the metal cover together.
- This plastic gasket is placed in between, and therefore this.
- This is the all the components which we saw just now.
- They are all placed like this inside the battery.
- Then we have the anode terminal and the cathode terminal, and the in between there is a.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 10:00–14:59

gasket which is for preventing the anode and cathode from coming in direct contact with each other.

Key details

- Plus 2OH minus is giving you ZnO plus H2O plus 2 electrons.
- We should put the two electrons here because we are talking of the anodic reaction.
- Oxygen has to undergo reduction.
- The reaction is half O2 plus H2O plus 2 electron is giving you 2OH minus.
- Plus half O2 is giving you ZnO.
- Zinc is reacting with oxygen to give you Zn oxide.
- But if you look at the schematic diagram which when I talked about battery we I showed you the diagram where we talked about working of the battery we have the anode we have the cathode and we have the electrolyte and the separator.
- Is the anode material, and at the cathode we have this catalyst which is placed here, and we have this.
- Holes through which oxygen can enter.
- This is just a schematic representation and not the actual representation because the electrolyte is not.
- Present in bulk like this, but it is present in the separator only.

- The separator soaked in electrolyte.
- is produced here and this move towards the moves towards the zinc electrode and then we have this reaction happen at the animal.
- So electrons are being produced here, they are getting absorbed here and therefore we get power from this battery.
- Course batteries all batteries are galvanic cells, so this is how this the battery works.
- This is about the working of the battery, and we have.
- EMF is 1.4 volts.
- The separator soaked in electrolyte.
- The reason for that, if you look at the final reaction, zinc plus half O2 is giving you Z and O.
- OH minus is not figuring in the final reaction.
- That means the amount of OH minus used.
- is not going to affect the working of the battery.
- Whatever is produced at the cathode is being used at the anode.
- Just a small amount of QH just to facilitate the movement of the ions is what is required.
- Hence, we need not use a large amount of electrolyte and the separator soaked in electrolyte is sufficient, a small amount of electrolyte.
- Which is soaked inside the separator is enough for the battery to function and therefore.
- One important characteristic of this battery is that these batteries can be miniaturized.
- Because the amount of electrolyte is not important.
- If you look at lead acid battery, we know that in the final equation we have H2SO4.
- Of H2SO4 matters, and therefore, lead acid batteries always come in large sizes.

Example or application from the lecture

- Let us look at the working of this battery.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 15:00–19:59

You have to make them large, it cannot be made small.

Key details

- Whereas here OH minus the electrolyte is not there in the final equation.
- So that means final reaction.
- So amount of electrolyte can be very small, and therefore these batteries can be made into very small sizes.
- The word the term we use to describe that is that these batteries can be miniaturized.
- high energy density as I explained right in the beginning and that is because air which is the cathode active material is not getting stored inside the battery it is taken directly from the atmosphere therefore the weight of the battery is sufficiently reduced and therefore this result in a very high energy density of the zinc air battery.
- Second, it has a very long shelf life.

- We cannot say unlimited shelf life because there could be some other reasons like corrosion, etc., which can happen when you store the battery for a long time.
- But nevertheless, the shelf life is very long because as long as the battery is kept sealed.
- The air cannot enter.
- So that means oxygen cannot enter the holes which are there on the battery.
- If they are sealed, oxygen cannot enter the battery and therefore the reaction cannot happen.
- Therefore, you can store these batteries for a very long time and therefore they are.
- They have very long shelf life.
- They also do not pose any ecological problems.
- There is a zinc and oxygen which are being used in the battery, and it is a low cost battery because zinc.
- oxygen is anyway taken from directly from the atmosphere and zinc when used in manufacture in mass scale the cost becomes very low so it's a low cost battery.
- And the second important point which we need to know is that along with air, when we open the battery, then there is carbon dioxide also in the atmosphere.
- And since we have used alkali here as the electrolyte, what the electrolyte is KOH, therefore, there could be a reaction between this carbon dioxide and KOH, which is going to result in formation of K2CO3.
- And that is going to block the pathways and therefore reduce the efficiency of this battery.
- You can see the first application is in hearing aids.
- Because these batteries, as I told you just now, can be miniaturized, they can be made into very small sizes.
- So this is about the Zincare battery, and in the next class we are going to look at the batteries which are used profusely in the present day, and that is lithium batteries.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

6 Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1)

Study focus: Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000086 · EV0000087

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module 5 on energy storage devices on fuel cells.

Key details

- In a thermal power plant, we know that combus of fuel is done and power is generated from there.
- But this happens in several steps.
- When the fuel burns, there is heat generated.
- This thermal energy is used to run turbines.
- These turbines now are used to generate power.
- To electrical energy, but at each step there is a loss of efficiency.
- The idea in a fuel cell is can we convert the chemical energy directly into the electrical energy?
- If that is done.
- We will definitely have a better efficiency, and that is exactly what is done in a fuel cell.
- The chemical energy is directly converted into electrical energy, and we find that the efficiency is about 75 to 80 percent.
- Hydrogen is the fuel and oxygen is the oxidant is done in the form of galvanic cell.
- We can see very clear hydrogen is getting oxidized to water and oxygen is getting reduced to water.
- So therefore, we know that combustion of a fuel it is a redox reaction.
- And we know in a galvanic cell we can use this redox reaction.
- How we can make the reaction happen in two different compartments, meaning that we will not allow the hydrogen and oxygen to come in contact, but we will allow the hydrogen to undergo oxidation at the anode and oxygen to undergo reduction at the cathode.
- Since hydrogen is undergoing oxidation.
- At the anode electrons will be generated and these electrons are going to move through the external wire towards the cathode where there is oxygen and that is going to absorb the electrons and undergo reduction.
- So whatever is the chemical energy of the fuel oxygen system is getting directly converted into electrical energy by means of a redox reaction.
- And they are they undergo reaction, and that is how we get power from the battery.
- But a fuel cell, though it is also a galvanic cell, is different from a battery.
- Because the fuel and the oxidant, that is the electroactive material, have to be supplied.
- Continuously from outside as at the respective electrodes.
- Fuel cell, the fuel and the oxidizing agents have to be continuously supplied.
- They will be able to generate power only as long as the electroactive material is supplied at the anode and the cathode.

Example or application from the lecture

- The combustion of fuel, that is, say for example, hydrogen and oxygen.
- Let us check whether the combustion of fuel is a redox reaction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let us look at the construction.

Key details

- What would be what would the fuel cell look like?
- We understood the concept.
- Definitely, because we are talking of an cathode, we need two electrodes.
- But apart from the electrodes, one very important part component of the battery is going to be the catalyst.
- Because the reactions which take place during the fuel oxidation or the oxygen getting reduced or the oxygen getting reduced, they are very slow reactions.
- So the catalyst also is very important for the fuel cell because the reactions are very slow.
- And apart from that of course we need the electrolyte like any galvanic cell.
- They are not being stored inside the.
- cell in a fuel cell, but they have to be supplied externally.
- The electrolyte and the electroactive material which are being supplied from outside.
- Separating the fuel cell into three compartments.
- They make it into three compartments.
- The anode compartment, the cathode compartment, and the electrolyte.
- These are the.
- Two electrodes which are present, of course, and we know that the catalyst has to be present on the electrode.
- We have on this side we have the oxidant coming inside, and here we have the excess oxidant coming out of the cell, and in between we have the electrolyte.
- is the anode because the fuel undergoes oxidation and since the oxidant is coming in from here, this I would call it as the cathode.
- and in between we have the electrolyte.
- Which we are getting from the battery.
- We always start with the anode, like in any galvanic cell.
- Then we have the electrode at the cathode, and then we have the oxidant.
- We know that at the anode, the fuel is undergoing oxidation, so electrons are being generated, and when the fuel undergoes oxidation, obviously the product is going to be the oxidation product of the fuel.
- At the cathode we have the oxidant which is taking up the electrons and it will give rise to the reduction product whatever is the.

Example or application from the lecture

- Let us see what do we need to make a fuel cell.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 10:00–14:59

oxygen corresponding reduction product is going to be formed at the cathode.

Key details

- Of course we can have a variety of fuels which we use.
- And also a variety of oxidants that can be used.
- Hydrogen is the most popular.
- fuel which is used.
- And then we also have carbon monoxide, which is also good fuel, which will get oxidized to carbon dioxide.
- And then in we have methanol, which is catching up these days, and methanol oxygen fuel cell is also becoming very popular.
- And in place of methanol, we can have ethical.
- Then there is formaldehyde and hydrazine.
- They can undergo oxidation very easily.
- Of course, we understand the efficiency is very high, much higher than what we see in the thermal power station.
- The efficiency is about 50 to 80 percent in the case of a fuel cell, and most importantly, they are eco-friendly because the products which are formed are not toxic.
- If we are using methanol we are going to get carbon dioxide and water.
- They undergo combustion, the which are used in vehicles, we know that there's a lot of pollution happening due to the.
- And then another thing, another important fact is that the advantage is that it they have a very silent operation.
- Unlike in a thermal power plant, there is there is a there is no noise when such a power generation is happening in the fuel cell.
- But there are certain disadvantages also.
- The cost of power is very high.
- Because we have to go for expensive electrodes and also catalysts.
- More important are the catalysts because they have to be present for the reaction to take place.
- And these catalysts, electrocatalysts especially, are expensive.
- And one more important fact is that fuels which are used in the form of gases.
- and oxygen they have to be stored in tanks under high pressure.
- Hydrogen is a very good fuel, but storing hydrogen is the biggest challenge for using the fuel cells.
- What are the applications?
- Yes, space exploration.
- All of us are aware that it started a long time back and NASA uses these fuel cells right from the Apollo space program.
- powering the space vehicles.
- They act as auxiliary power generators in space vehicles and also vehicle traction.
- for cars, buses and commercial vehicles, they are looked upon as the next generation power generators for cars, buses and commercial vehicles.
- So vehicle traction, traction means moving, when you're you using not fuel, but you're using the battery or the galvanic cell is being used, in this case the fuel cell is being used to power the motor to run the machine.

- Stationary applications have been done to a great extent, but still mobile applications are still being researched.
- So if that comes then we are going to be very well placed as far as power powering the vehicles is concerned because we know that the non-renewable resources like petroleum, etcetera, the fossil fuels.

Example or application from the lecture

- Products being given out into the atmosphere.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 15:00–19:59

They are depleting fast, so if we could get these fuel cells to power the cars, buses and commercial vehicles and also for large scale power generation, it is going to be very useful in future.

Key details

- Any efficiency is always measured by what is the output work done by the input, input heat in this case.
- Gives free energy because we know that delta G is equal to minus N FE.
- So we know that the electrical output which you get, the electrical energy which we get from the battery is because of the Gibbs free energy.
- And what is the heat input in this case?
- It is the enthalpy change which happens.
- During the cell reaction.
- Which reaction are we talking about?
- We are talking about the combustion of the fuel.
- So whenever the fuel burns in oxygen or any other oxidant, then the heat which is released is the is the heat input that is because the chemical energy is what we are inputting in using as input in the fuel cell.
- So therefore we have.
- Heat input is enthalpy change of the cell reaction and the work output is the Gibbs free energy.
- That would be it's denoted by the symbol, this symbol percentage that is delta G by delta H.
- In the next class we will look at hydrogen, oxygen, alkaline fuel cells.

Example or application from the lecture

- Let us do efficiency of a fuel cell.
- We can say it is the electrical energy which we are electrical work which are which we are getting from the fuel cell, and that is given by the Gibbs free energy.
- The efficiency is given in the form of percentage.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

7 Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2)

Study focus: Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000102 · EV0000103

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module 5 on energy storage devices and we'll discuss about fuel cells.

Key details

- In today's class, we are going to learn about hydrogen oxygen alkaline fuel cell.
- The principle, construction and working, the advantages and the disadvantages.
- Alkaline fuel cells as the name suggests, the electrolyte which we are going to use is going to be an alkali.
- If you remember when we talked about fuel cells in general, we use a fuel at the anode and an oxidant at the cathode, and there is an electrolyte.
- These are low temperature fuel cells and they operate at about 100 degrees centigrade.
- Most common oxid which is used at the cathode is oxygen.
- Oxygen reduction is a very slow process.
- But if we use alkaline electrolytes,.
- As compared to acid electrolytes, it is found that the oxygen reduction is much more rapid in alkaline electrolytes.
- And because of that, we need not use noble metal catalysts.
- Oxygen reduction or hydrogen oxidation.
- Hydrogen oxidation which happens at the anode or oxygen reduction which happens at the cathode being slow processes we need to use catalysts.
- But because it is faster, the oxygen reduction is faster in the alkaline electrolyte as compared to acid electrolyte, we can use non noble metal electrocatalysts.
- One problem with alkaline fuel cells is that carbon containing fuels cannot be used as when they undergo oxidation, carbon dioxide is formed as a product.
- This K2CO3 is going to block the electrodes, the electrode pathways, and therefore this reduces the efficiency of the cell.
- The reaction would be carbon dioxide, which is formed by the due to the oxidation of carbon containing fuels with KOH is going to give you K2 CO3 plus H two O.
- used and has many applications and the electrolyte obviously is going to be an alkali.

- At the anode it will be hydrogen which is going to be used.
- At the cathode we will use oxygen and the electrolyte is going to be an alkali.
- The oxidant which we use is oxygen.
- and the electrolyte is KOH that is 30 to 45% of KOH is used.
- But one thing we need to note here that the KOH used has to be warm.
- We'll discuss the reason in the upcoming slides.
- These are the electroactive materials.
- Hydrogen and oxygen are the electroactive materials, but we also need electrodes.

Example or application from the lecture

- Let us discuss the construction of such a fuel cell.
- Now we will take the example of hydrogen oxygen fuel cell, which is the most common type of alkaline fuel cell.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let us look at the diagram.

Key details

- Hydrogen because hydrogen has to undergo oxidation and at the anode the process is always oxidation.
- Carbon electrode is impregnated with silver.
- One compartment is the anodic compartment where we let in the fuel.
- And we also have a exit for the excess fuel and on the cathode side that is the other compartment where we have the cathode here is the entry for air or oxygen in this case and then there is also an exit point for the excess air or oxygen.
- and in the center we have the electrolyte that is the KOH which is being used.
- And apart from that we have the electrodes which are shown here that is the anode and this side is the cathode.
- When we say working, we are basically going to understand the reactions which are taking place at the anode and cathode and the overall reaction.
- And then at the cathode, these electrons are getting absorbed.
- Overall reaction when you add up these two reactions the half cell reactions they are known as the anode and the cathode, we find that it is hydrogen plus half O2 is giving you water.
- But what we have done is we have separated the oxidation and the reduction reaction into separate compartments.
- These electrons move through the external wire and they are taken up at the cathode because of the reduction of oxygen happening here.
- So this way by the separating the reactions into anode and cathode just like we do for any galvanic cell, we have been able to have the stream of electrons flowing and therefore we can generate power from this cell.
- Hence, it is very important that the anode has to be porous in nature.

Example or application from the lecture

- Let us understand the working of such a cell.
- OH minus and it is going to form water and two electrons are given out here because it's a oxidation reaction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 10:00–14:59

Expedite the reaction.

Key details

- In some cells we find that there is a vent provided over here and from here the water escapes.
- Whenever we make any fuel cells it is very essential that the electrodes have to be porous in nature because they have to absorb the fuel the oxidant and also the catalyst is present on the electrode.
- So therefore the hydrogen oxygen is getting ads at the cathode surface and it gets reduced to OH minus ions.
- Therefore, what we do is we operate the cell at 100 degrees so that the water which is formed from KOH can be it is going to escape as steam.
- Because this water which is getting formed is obviously going to dilute the KH and we have used a particular concentration of KH.
- And if the KVH gets very diluted, then the efficiency of the reaction is going to fall down.
- Therefore, we make sure that this excess water.
- Which is getting formed continuously is removed from the system.
- Hence, it is very important that water is allowed to go out of the system, and that is why we operate the cell at about 100 degrees centigrade, or in other words,.
- Why we use warm KH so that water escapes as steam and we maintain the concentration of KVH which is required for that for the reaction.
- And one more interesting fact which you must be familiar with is that during the Apollos phase program the water which was coming out of this fuel cell was collected and the astronauts were using this water for drinking.
- Now the EMF which is generated because we are using this fuel cell basically to generate power.
- Coming to advantages of this fuel cell, it operates at a low temperature.
- So because of that, corrosion of equipment which is being used in the fuel cell happens to a very small extent as compared to those fuel cells which are operate at a very high temperature.
- And alkali which is used as the electrolyte is allowing us to use a non-noble metal catalyst and because of that what happens is the cell is now less expensive.
- So we can afford to use non-mobile metal catalysts here and therefore the it will work out to be much less expensive as compared to other fuel cells.
- And there are certain disadvantages that the reactants must be free from carbon.

- We also saw this point that if we have carbon containing fuels, then they will undergo oxidation to form carbon dioxide, and this reacts with alkali.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 15:00–19:59

And it forms carbonates, which is going to reduce the efficiency of the cell.

Key details

- which have which are used has to be free of carbon compounds.
- And other problem is that since we are using electrolyte which is a liquid, then that could pose some handling problems.
- And depending on the application where we are going to use this fuel cell, we will decide which type of fuel cell is suitable or whether this fuel cell is suitable for that particular application.
- So in the next class we are going to talk about methanol oxygen fuel cell where we are going to use a polymer membrane as the electrolyte.

Example or application from the lecture

- So it also means that the fuel and the oxidant, suppose we are talking of hydrogen and oxygen, then fuel and oxidant.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

8 CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells

Study focus: CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000118 · EV0000119

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll continue with module 5 on energy storage devices that is on fuel cells.

Key details

- This fuel cell is also known as proton exchange membrane fuel cells, thus telling us that the polymer which we are going to use is capable of exchanging protons.
- Why is methanol fuel cell so popular these days.
- If you remember in the hydrogen oxygen fuel cell I mentioned that storage and transport of hydrogen is going to be difficult because it is in the form of a gas.
- But methanol being a liquid.
- We can easily transport and store the methanol, and hence there is a possibility that we can use it for portable devices.
- This fuel cell has low weight and volume.
- Low weight and volume because we are we are using polymer as the electrolyte and polymers are lightweight.
- So therefore this cell will have very low weight and volume and therefore.
- What is the good characteristic of this fuel cell?
- That it is going to have very high energy density.
- The weight has reduced, therefore the energy density will be very high.
- But there is one problem we face with this fuel cell, that we have to use only noble metal as catalyst, and usually it is platinum which is employed as the catalyst in these cells.
- And one more problem which we face that carbon monoxide.
- If it is present, then it is going to poison the catalyst.
- We are using platinum as the catalyst.
- It is going to poison the catalyst, and therefore pure fuel or oxidant only should be used.
- The electrode is a porous carbon electrode containing platinum.
- Cathode is also a porous carbon electrode containing platinum.
- Just now I mentioned that noble metal only should be used, but then we always have to have a porous electrode.
- because all reactions are going to take place on the electrode.
- The catalyst is present there, the electrolyte is in contact, and also the fuel or the oxidant is going to come in contact with the electrode.
- It is a solution of methanol.
- We never use pure methanol.
- We are going to use a solution of methanol and the oxidant is oxygen gas.
- of methanol and the oxidant is a oxygen is the oxygen gas and the electrolyte as we are studying now it is a proton conducting polymer membrane now any electrolyte for that matter.
- What should be the criteria that it has to be an electronic insulator?
- What does it mean?
- That it is not going to allow the electrons to move through itself that is the electrolyte should be an electronic insulator.
- Yes, we are using a polymer here, so this membrane is an electronic insulator, but at the same time we want the hydrogen ions to move.
- That's why we have chosen a proton exchange membrane.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 05:00–09:59

Excellent conductor of the hydrogen ions.

Key details

- Electrolyte is usually the most common electrolyte used is a fluorocarbon backbone.
- It's similar to Teflon.
- Teflon how does it look?
- It's a CF2, CF2, CF2.
- You have so many units of CF2.
- Which are bound together and that is Teflon.
- Because we know that sulfonic acid has.
- The SO3 minus and H plus it can dissociate in water as SO3 minus and H plus so that this proton now is free to move.
- It becomes like this in the presence of water.
- The H plus ions obviously are going to be able to migrate through when the membrane is hydrated.
- So when because when it is hydrated, you will be able to get the H plus ions to move through the electrolyte.
- The most common one in any protonic stain membrane electrolyte fuel cell or we can call it as a polymer electrolyte fuel cell.
- It's always napion.
- It is a very popular polymer which is used in these fuel cells and that's because it has very high proton conductivity that's what we want the diode should move where very easily through the membrane and it has good mechanical strength and it also has good chemical stability.
- As I said, there's a backbone of CF2, and then that is the Teflon backbone, and then here we see the SO3H groups which have been appended onto this backbone.
- is acquebn which also has a similar structure just that the CF2 branching I mean the CF2 chain has been reduced.
- Here it is a longer chain and here you can see it's a smaller chain.
- Nevertheless Nepheon happens to be the most popular membrane which is being used.
- But when we come to DMFC, what is DMFC?
- Direct methanol fuel cells.
- That is what we are studying now.
- Nafheon is not suitable for such a fuel cell.
- where we are using methanol and it's a diet methanol fuel cell.
- Because along with H plus ion movement naphion is highly permeable to methanol.
- What happens because of that?
- It's allowing the methanol also to move through the membrane and therefore a phenomenon known as methanol crossover happens.
- Here you can see the methanol diffuses through the membrane to the cathode.
- So instead of reacting at the anode undergoing oxidation at the anode we find that methanol just crosses over to the cathode and there it is going to combine with the oxygen which is

not what we want because if we want to generate power from the cell then oxidation of methanol has to happen at the anode but.
- But unfortunately, when we use Nafion, it is found that this methanol crossover happens.
- So, therefore, that will result in what you will get lower voltage because less number of electrons are being generated.

Example or application from the lecture

- Let us see.
- Ether ketone the structure is given here.

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 10:00–14:59

Many new membranes are there.

Key details

- Sulfonated polyether ether ketone is used as the membrane.
- We can see the picture here that.
- We have methanol and water which is coming from here.
- And what would be the reaction?
- We are letting in oxygen.
- From the anode to the cathode, and it is going to take up the electrons which are coming through the external circuit and form water.
- exactly what is happening here.
- A weak solution of methanol is fed.
- One reason for a weak solution being used is that methanol crossover is minimum when we use a weaker solution.
- And then the common operating temperatures, as already mentioned, is 60 to 90 degrees centigrade, and the oxidation of methanol on a catalyst layer takes place, and carbon dioxide is.
- What is your electrolyte?
- It is a polymer electrolyte membrane or a proton exchange membrane.
- If the membrane is hydrated the H plus ions will.
- Just hop through the electrolyte and reach the cathode.
- Oxygen reacts with this H plus ions as we saw in the reaction, and then it is going to give out.
- Anode the water is being consumed at the anode because we said a weak solution of methanol is being fed over there.
- Travel through an external circuit.
- Water management in this fuel cell is very important and the temperature has to be maintained 60 to 90 degrees.
- We cannot have a higher temperature, we cannot have a lower temperature.

- Unless it is hydrated you cannot have the H plus ion moving through because only then the SO3H is going to dissociate.
- and water which is produced during the reaction it has to be removed from the cathode very frequently.

Example or application from the lecture

- One example is SPEEK.
- with water reacts with water a solution of methanol we have used and that is going to generate the H plus and we are interested in these H plus ions and electrons are given out plus CO2.
- Being given out as we saw in the reaction, we get carbon dioxide as a product, and at the same time, protons are being generated, and these protons now are going to migrate through the electrolyte.

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EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 15:00–19:59

This is called water management.

Key details

- Now when we have high temperature, what is the problem if we use high temperature what will happen the polymer may get dehydrated and therefore H plus conductivity cannot take place and also the polymer may degrade and crack resulting in short circuiting.
- And another reason for this, another fact which we have to know is that.
- Neither can low temperatures be used because this will result in flooding the cell because water will not be removed if you keep it at a very low temperature.
- Water will not be removed and this will result in flooding of the cell and that reduces the efficiency of the cell.
- And also because the temperature is very low, we will require a higher loading of catalyst.
- We are using platinum as the catalyst.
- we have we it will be very expensive.
- Neither can we use a high temperature, neither can we use a low temperature.
- What are the advantages of this fuel cell?
- So as mentioned it is it has a very high energy density because of the low weight of the polymer.
- And then it has a long life because we are using work making it work at a very low temperature and therefore the with low temperature it will start very quickly.
- first of all, and it also results in less wear and tear of the system components and therefore a better durability.
- No free corrosive liquid is being used in the cell because the electrolyte is a polymer.
- Therefore, there is no free corrosive liquid which is being used in the cell.
- Noble metal catalysts have to be used and the electrolyte is also expensive.
- We are using special electrolytes, special proton exchange membranes have to be used.

- And if we have CO then carbon monoxide if at all present it is going to poison the platinum catalyst and hence pure reactant should be.
- And management of carbon dioxide, which is created at the anode and water management pose problems.
- As I mentioned that water shouldn't be present in a right amount, when in the right amount and it should not be in excess.
- just now we discussed water management.
- In the next class, we are going to look at another fuel cell that is a solid oxide fuel cell where we are going to use carbon monoxide as the fuel.
- fuel cell using solid oxide as the electrolyte.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

9 Supercapacitors , Ragone plot

Study focus: Supercapacitors , Ragone plot. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000134 · EV0000135

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module five on energy storage devices on supercapacitors and this will be the last class of this module.

Key details

- When we use a power source, the negative terminal is attached to one of the plates so that gets the negative charge and the positive terminal when it is attached to the other plate that plate gets the positive charge.
- The charge stored in a capacitor is directly proportional to the voltage developed across the capacitor.
- Then C is known as the capacitance and the unit used is ferrats.
- will decide the capacitance of the capacitance because capacitance is directly proportional to the area of the plate as you can see in the picture here.
- This is the area of the plate and it is inversely proportional to the distance between the two plates.
- What are supercapacitors then?
- They are also known as ultra capacitors or electrochemical double layer capacitors, E, D, and C.
- you can understand that the capacitance has gone up to such a high extent and they also charge and discharge very quickly.

- This is for any capacitor and of course it is applicable to supercapacitor also that they charge and discharge very quickly.
- Why is the capacity so much higher than that of a capacitor?
- We have electrodes, we have the electrolyte and we have a separator.
- Very important high surface area materials.
- The separator is an iron permeable separator and it's.
- As in batteries, it is placed between the electrodes, and what is the role of a separator?
- It is always to prevent electrical contact between the anode and the cathode, but it is going to allow the ions to.

Example or application from the lecture

- Before we talk about supercapacitors, let us understand what is a capacitor and how is a supercapacitor different from a capacitor.
- If we know that capacitor is usually in the capacitance is given in microfarads, but for a supercapacitor it will be listed in thousands of ferrats.
- Let us look at the construction.

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EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 05:00–09:59

move from the one electrode to the electrode to the other electrode and the example for this would be porous polypropylene.

Key details

- How exactly are these components placed?
- You can see in this picture.
- The separator is sandwiched between the electrodes.
- The high surface area electrode which are highly porous.
- And then this whole system is now placed in a case and then electrolyte is poured over it.
- It goes through all this porous electrodes which we have placed over here.
- And then of course the electrodes we need.
- Just like in a battery, we need current collectors.
- When a potential is applied, so the positive electrode, one of the electrode obviously will become positive based on the terminal of the external source.
- And similarly, the negative electrode which is connected to the negative terminal is going to become is going to get attract the positive ions.
- So and it and this double layer now because you made a porous electrode it is formed at the entire electrode electrolyte interface and plus the charge separation is just nanometer range.
- What is happening here?
- Only absorption and desorption.
- They get they form the double layer and then when you are discharging then these ions are getting away from the electrode surface.
- that is happening here.

- So because of this.
- We find that this supercapacitor has many advantages, which we will discuss now.
- It is just this absorption and desorption which happens during the charging and the disorganization.
- charging process now what has happened the distance between the charge lay has become into the nano range as I explained just now so D is very small and.
- Also we used porous electrodes.
- So because of the porous electrode, what do we get?
- We have a very high surface area.
- The moment you make a material porous, we know that it is going to get in get a very high surface area.

Example or application from the lecture

- Let us look at the working.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 10:00–14:59

We know about capacitance C is proportional to A by D.

Key details

- If the area is very high and the D is very small, in fact here it's in the nanometer scale.
- Now this formation of electric double layer of course takes place at both the electrode electrolyte interface and therefore at each of these electrodes we have the capacitance.
- And the total capacitance would be 1 by C T is equal to 1 over C plus 1 over C minus.
- What are the advantages of these supercapacitors?
- They charge very, very quickly.
- Because it just involves the movement of these ions towards the electrodes and forming of the double layer.
- And very high.
- power density because power density depends on how fast the energy is getting discharged and we know that they can discharge also very quickly because there is no redox reaction that is taking place.
- It's only desorption of ions that is required.
- And of course high cycle life they can be cycled millions of times as compared to your battery, which the best batteries have about five hundred to thousand is very high cycle life, but here we have device which can be cycled several time because again there is no redox reaction that has to take place.
- It is just the absorption and desorption of ions which can happen as many times like we can say it's almost unlimited cycle life.
- And yes, compared to batteries, they are much safer as the internal resistance is very low, and when the internal resistance is low, we know that the heating rate is also going to be low.

- Then there are certain disadvantages also.
- What is it that they have very low energy density.
- Low energy density.
- What is the reason that it the reason is they discharge very quickly, so obviously they cannot retain or hold the charge for a long time.
- So energy density is low, and of course, high self-dcharge because it just involves absorption and disorption of ions.
- High self discharge.
- Linear discharge voltage.
- If you remember for a battery when we draw voltage versus time, the line looks like this the graph looks like this what happens it remains steady for a long time and then there is a drop in the voltage when we say the battery is dead.
- Whereas for a supercapacitor, the drop is linear.
- So therefore the volta is going to drop very quickly with time, whereas in a battery we find that we can get a steady voltage for a reasonable amount of time.
- This supercapacitor because of the electrodes which are you being used, very highly porous electrodes have to be used.
- They are very expensive.
- And then the power is available for a very short duration because they discharge very quickly.
- Where would such.
- Batteries be useful or supercapacitors be useful.
- They are used for memory backup.
- Then hybrid cars for start stop applications in flash photography where you need high power for a very short period of time.
- flashlights, portable media players, and also as an intermediate energy storage for FM radios, cell phones and emergency kits.
- We have talked about different types of batteries, we talked about fuel cells, we talked about supercapacitor.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 15:00–19:59

Energy storage devices comparison compare the performance of these and that can be done when we plot energy density.

Key details

- Energy density versus power density for a whole range of energy storage devices and why is it possible to use to plot all these energy different energy storage devices?
- That's because it's a double log chart.
- The reason that they discharge very quickly, so they can give you a very high power density.
- They have very high power density, but they don't retain the charge for a long time, hence they have low energy density.

- The reason being that there is no redox reaction.
- is simple absorption and desorption of ions that has to take place and therefore they charge as well as discharge very quickly.
- We also studied about fuel cells now fuel cells have very high energy density but low spow density.
- The reactions have to take place, and the reactions are usually very slow.
- If you remember, we always use catalysts.
- So therefore the reactions are slow, and hence the few if our fuel cells the power density is going to be very low, whereas the energy density is high because we can supply the fuel and the energy and it can work as long as you supply the fuel so the energy density is very high.
- Batteries of course come somewhere in between they are occupying this position, solid acid battery, nickel cadmium battery, lithium ion battery, lithium ion battery.
- Has an advantage because of lithium being used, which is a lightweight metal.
- And of course, when you talk of combustion engines, we know that we can keep on supplying the gasoline and therefore the energy density is going to be high and the reactions are very quick where we know that chain reactions happen.
- So therefore the power density is also very high.
- So fuel cells will have high energy density and whereas they have low power density batteries will have moderate energy density and power density ultra capacitors or supercapacitors they can deliver very high power therefore power density is very high whereas the.
- Energy density is going to be low for a ultra capacitor or a supercapacitor.
- So in the this is the end of this module, and in the next class, we are going to talk about.
- Then next topic and that is corrosion.

Example or application from the lecture

- that is given in watt hour per kg versus the power density that is watt per kg of a particular device and such a plot is known as a raguni plot.
- Let us have a look at this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

10 Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2))

Study focus: Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000150 · EV0000151

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module 4 on electrochemical equilibria.

Key details

- We look at the construction, the working and how to use the glass electrode to determine pH of a solution.
- It's an ion selective electrode which responds to H plus ions.
- And therefore we can say that it is pH sensitive.
- So like any other ion selective electrode, there has to be a membrane which is going to be able to exchange these H plus ions in this case because we want to determine pH of the solution and here it is the glass membrane.
- We have a bulb.
- Now, this bulb is the most important part of the glass electrode because this is made of a very thin membrane of glass, and this glass is of a specific type, and that is corning 015.
- It is used to make this glass.
- The bulb the glass bulb is made up of coning 015, and this is a very thin membrane.
- We need a reference solution.
- And that is.
- HCl of 0.1 molar concentration.
- And in and in this solution, there is the internal reference electrode that is the silver chloride.
- And then we have the internal reference solution of HCl of 0.1 molar concentration, and then we have the internal reference electrode, which is the silver chloride electrode.
- We have to place this glass electrode in a solution which contains H plus, the concentration of which is not known to us, and therefore using this glass electrode we are going to find out.
- the concentration of the H plus ion in the analy solution.
- If you write only this, then that would mean that we are talking about the silver chloride electrode.
- But since glass electrode and we know it is only the glass which is capable of exchanging these H plus ions and not the silver chloride electrode.
- Therefore, we have to mention the glass here.
- That is the electrode representation.
- This is the schematic representation, just like we have for the ion selective electrode, we have the membrane here, and on one side of the membrane we have the reference solution.
- In this case, it is 0.1 molar HCl.
- That is H plus solution plus Na plus glass.

Example or application from the lecture

- And in today's class we are going to study the glass electrode, which is an example for ion selective electrode.
- Let us look at the construction of this glass electrode.
- Let us look at the working of this glass electrode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 05:00–09:59

If you look at the structure happens to be Na plus Gl minus, and that is in equilibrium with Na plus solution.

Key details

- Plus h plus gl minus.
- What can you understand from this reaction?
- That NA plus Gl minus, which is glass, is being able to exchange this H plus.
- Plus goes into the solution.
- This is the exchange reaction which is taking place at the membrane, at the surface of the why is this possible?
- If you look at the structure of glass, stru glass is actually a three-dimensional network of silica tetra hetra.
- And since an open network structure we have this interstitial space which are filled with cations.
- It is going to depend on how the glasses.
- Is made if it's sodium silicate, we are going to have sodium ice.
- If it's alumino silicate, we can have aluminium ice, calcium silicate, calcium ice.
- If it's alumino, potassium silicate, we'll have both aluminum as well as K plus ice.
- This membrane comes in contact with the solution containing H plus ions.
- What happens is.
- The outer lay of this glass membrane are going to swell up.
- When they absorb this solution and they swell up.
- These H plus ions will go in and the sodium ions will come out.
- But mostly the divalent and the trivalent are held very tight.
- So therefore, it's only the monovalent ions which are found to be mobile.
- The H plus ion now you can see in the picture over here, since it is acidic medium, we find that H plus ion moves into this gel layer and the NA plus ion is going to come into the solution.
- What do we mean by exchange of these ions?
- From the solution, and based on the concentration of H plus ion, a potential is going to be developed at the membrane.
- At the membrane, we have a potential which is getting formed.
- Get an expression for the potential of the glass electrode.
- where we know it is 0.1 molar HCl.
- Again, this is internal reference electrode that is AG ADCl electrode.
- On the other side of the membrane, we have the analyte solution where we have H plus concentration is equal to C1.
- surface and this membrane surface are in contact with different concentration of H plus ions.
- The concentration C2, the outer membrane is in contact with the concentration C1.

- Now, because of this, at the outer membrane we get a potential E1, at the inner membrane we get a potential E2.
- At the boundary now, therefore, at the membrane boundary, we have a difference in the potential that is E1 minus E2.
- Because the membrane.
- However thin it is, it has an inner surface and it has an outer surface.
- The inner surface is in contact with one concentration of H plus, the outer surface is in contact with a different concentration of H plus.

Example or application from the lecture

- First let us understand that.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

Core explanation · 10:00–14:59

Now C2 is a constant because it is an internal reference solution.

Key details

- We already know it is 0.1 molar H Cl.
- Say the constant is L dash plus 2.303 RT over N F log of C1.
- It is minus of 2.303 R T over N F log of C2.
- nine one over n log of h plus now since.
- For H plus I n is equal to 1.
- Your expression now for the boundary potential we got L dash minus 0.0591 p h.
- We had the boundary potential plus the potential due to the internal reference electrode, which has been placed there to provide the electrical contact.
- Apart from this boundary potential and the internal reference electrode potential, we have a third potential coming up that is known as the asymmetric potential.
- So, therefore, the sum of boundary potential, reference electrode potential, internal reference electrode potential, and the asymmetric potential.
- Potential will give us the total glass electrode potential.
- We know that the glass bulb is curved, so the radius of curvature itself is different for the inner and the outer membrane.
- So what happens because of that?
- There is a small potential.
- Which is known as asymmetry potential.
- outside the analyte solution as the same concentration as the reference solution, we should not be able to measure any other potential except the potential due to the internal reference electrode.
- But we find that we even though both the concentrations are same, there exists a very small potential which is known.
- As the asymmetric potential, and that is due to the membrane itself, and it is.
- We get this expression.

- are all constants.
- So here we have E0 G is equal to L dash plus E reference plus the E asymmetric, and therefore we get the final expression for the glass electrode that is EG is equal to e naught g minus 0.0591 ph so if I know if I can know what is EG, E0 G being constant, I can.
- Find out what is the pH of the solution.
- But since this is a single electrode, we know that single electrode potentials cannot be measured.
- Therefore, we need to make a combination, and now what we have is we have to combine it with the external reference solution.
- solution using a solid bridge.
- You can see here this is the calomel electrode, which is the external reference electrode, and this is the glass electrode.

Example or application from the lecture

- Let us understand this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 15:00–19:59

cell representation for this combination.

Key details

- And why are we doing this?
- Because we want to see how to use the glass electrode to determine pH of the solution.
- We have the.
- External reference electrode that is the calomide electrode HE2Cl2 Cl minus.
- Through the solid here, through the porous grid here, it is in contact with the analy solution.
- That in turn is in contact with the solution, that is the internal reference solution, and that is in contact with the silver superchloride electrode.
- We have galomel electrode which is diped in analyte that is in contact with the glass which is in contact with 0.1 normal HCl or 0.1 molar HCl and that is in contact with the silver chloride and then silver.
- Using the glass electrode, how are we determining the pH?
- We know that we have made a combination and we can measure the EMF of this cell.
- This can be measured potentiometrically.
- E right minus E left, so E glass minus E calomel.
- E glass just now we got this expression that it is E naught G minus 0.0591 PH and when we substitute we get this expression E naught G minus 0.0591 PH minus E calomel and if you rearrange we get the expression for pH.
- E naught G is fixed for a particular glass electrode and then using this expression I can determine the pH of the solution.
- All the glass electrodes.
- I mean what I mean is E0 G is not the same for all the glass electrode.

- Therefore, before using any glass electrode, I need to measure what is E0 G for that particular glass electrode.
- Using the same expression or the same setup, I can do it.
- Do I do now?
- I take a solution of known pH.
- If I know pH and I know what is e calomel, I can measure E cell, I can determine what is E0 G.
- So this is how we are going to determine E0 G first, and that can be used for the to determine the.
- further once we know what is E0 G, then we can determine pH of any solution.
- That is with the help of a buffer solution.
- We use a solution of known pH, usually a constant pH like buffer solution, and we find out E0 G and.
- And further, knowing ERG we can just measure the E cell and get the values of the various solution the hydrogen and concentration of different solution.
- It is used extensively in chemical, industrial, agricultural, and biological labs.
- What are the advantages?
- It can be used in oxidizing and reducing environments.
- Which was not possible with the help of hydrogen electrode.
- Hydrogen electrode can also be used to determine H plus, but it cannot be used in oxidizing and reducing environment environments.
- And metal ions, of course, we know it's an ion selective electrode.
- It is going to.
- be sensitive or detect only the H plus ions and not the other metal ions, and it does not get poison unlike the hydrogen electrode, and it can be used for very small volumes.
- And one of the most important.
- Advantage is that we get accurate results up to a pH of nine.
- However, beyond the pH of 9, we need special type of glass electrodes which can measure up to 14 pH with a good amount of accuracy.
- And they are of course very simple to operate and can be used with portable instruments.
- What are the disadvantages of glass electrode?
- And whenever there is a membrane, it offers very high resistance.
- The same thing happens with the glass electrode also.
- It is common for any ion selective electrode.
- potentiometers for EMF measurements and secondly the glass membrane which we are using is a very thin membrane so it is very delicate and hence has to be handled very carefully.

Example or application from the lecture

- Let us find an expression now.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 20:00–24:59

Just now I mentioned up to pH of 9 it gives very good accurate results, but beyond the pH of 9, we see some error in the normal in the glass electrode, and that electric is referred to as the alkaline error.

Key details

- reaction which normally happens at in when there is a glass electrode in contact with h plus ions.
- And if sodium ions are more in the solution, then the reverse reaction happens.
- It's an equilibrium.
- Going into that means the electrode becomes responsive to Na plus ions.
- Therefore, the pH now which you measure is going to be lower than the.
- So what you measure will be lower because now it is responding to NA plus ions also.
- So that is about the glass electrode and in the next class we are going to look at another ion selective electrode that is the fluoride ion selective electrode.

Example or application from the lecture

- The NA plus ion gets into the into the glass and the H plus ion is now given out.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

11 Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1)

Study focus: Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000170 · EV0000171

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module five on energy storage devices and will discuss oxygen sensor.

Key details

- Firstly, we will understand what is an oxygen sensor and look at some applications of oxygen sensor.
- and specifically how is an oxygen sensor used in automobiles the principle construction and working.
- It is an electronic device that measures the proportion of oxygen in the gas or liquid being analyzed.
- First and foremost.

- Divers use it to measure the partial pressure of oxygen in their breathing gas.
- We know that in the atmosphere we have about twenty percent of oxygen, which is the right proportion for us to breathe.
- There in the atmosphere.
- Hence monitoring the partial pressure of oxygen is very important, and therefore they use oxygen sensors.
- They are equipped, the deep sea divers gas cylinders carrying oxygen are equipped with oxygen sensors.
- Another application is when scientists use as probes to measure respiration or production of oxygen.
- Oxygen analyzers used in.
- Many applications such as anesthesia monitors and respirators have to use these oxy sensors to determine the partial pressure of oxygen which is being used in these medical appliances.
- It is also used to measure the exhaust gas concentrations of oxygen in IC engines, that is in internal combustion engines.
- Mentioned here, that is to measure the exhaust gas concentration of oxygen in IC engines.
- When fuel burns the ideal ratio for combustion of air is to fuel would be fourteen point seven is to one.
- Why do we need this ratio?
- Because when combustion is complete, we know that the there will be only carbon dioxide and water which are going to be let out into the environment.
- into the atmosphere.
- But if the ratio is not at the stechometric air fuel ratio, then there is going to be other gases which will be let out into the atmosphere.
- atmosphere which are harmful to us.
- And therefore, if we want pollution to be minimum, then.
- Th it is the stechometric air fuel ratio which should be used in the IC engines.
- And of course this ratio is for gasoline.
- ratio so we know that we require the air to fuel ratio should be 14.7 is to one.
- Now the oxygen sensor therefore allows the engine control system to maintain this ideal ratio across various engine operating systems.
- We know that when an automobile is moving then.

Example or application from the lecture

- Let's define what is an oxygen sensor.
- So first let us understand the principle based on which the oxygen sensor works and specifically in automobiles.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 05:00–09:59

The engine operating conditions keep changing based on whether it is moving slow, whether it is moving fast, whether it's going on a terrain which is not smooth.

Key details

- So every time the operating conditions are changing and the oxygen sensor therefore allows the engine control system to maintain the ideal ratio across all these conditions.
- It is going to compare the amount of oxygen in the exhaust to the amount of oxygen in the atmosphere.
- And we know that in the exhaust stream some amount of oxygen is going to be let out after the combustion reaction has happened.
- It is going to give an output voltage.
- Point two, then that represents a lean mixture.
- What do we mean by lean mixture?
- A lean mixture would be one which is lean in the fuel.
- That means that there is less of fuel and more of oxygen.
- Whereas if it is a rich mixture, rich mixture meaning it is rich in the fuel, there's lot of fuel and less of oxygen.
- The required amount of oxygen is not available and in that case the voltage is 0.8 volts.
- Now this is understandable because if it is a lean mixture that means there is more oxygen in the exhaust team.
- It is a rich mixture, there is going to be less oxygen in the exhaust stream, and when you compare it with the atmospheric oxygen, the difference is going to be higher.
- As the output voltage.
- The ideal set point would be around 0.45 volts.
- which it is receiving the signal voltage signal which it is receiving it is going to adjust the air fuel ratio back to the stechometric value.
- That means complete combustion.
- What happens when there is a lean mixture?
- When we have a lean mixture that means the there is less of fuel and more of oxygen.
- But something else happens here.
- The emissions now contain NOx gases like NO, NO2, etc.
- These NOx gases, if we want to get rid of them, because they also cause pollution, we know smog or acid rain is the result of these NOx emissions.
- But we are providing an oxidizing atmosphere because we have a lean mixture.
- There's a lot of oxygen and hence we find that NOPS emissions will increase if we have a lean mixture.
- Whereas if it is a rich mixture, that means the amount of oxygen is not sufficient to cause the combustion.

Example or application from the lecture

- Let us see further what does it do.
- We are just making sure that there is complete combustion and therefore only benign gases are let out into the atmosphere or less harmful gases are given out into the atmosphere.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 10:00–14:59

harmful gases which are going to be let out into the atmosphere and like carbon monoxide and then also some carbon particles will be there and some unburnt fuel.

Key details

- So we understand that neither should it be a lean mixture nor should it be a rich mixture, and therefore the engine needs to work at the stechometric value that is 14.7 is to one and the oxygen sensor is helping us to do maintain that stechometric value.
- The cell is a solid state electrochemical fuel cell.
- Just now we discussed about different types of fuel cells and in that one of them was the solid state fuel cell.
- Oxygen sensor is basically an electrochemical fuel cell which is going to use solid as the electrolyte and we also know that when we use solid as electrolyte then we need then the cell will operate at a high temperature.
- So it is going to operate only at higher because we are using solid as the electrolyte.
- The anode here is platinum.
- Since it's a fuel cell, we know that there is going to be an anode, there is going to be cathode, and we have the exhaust gas on one side and the reference air on the other side because we are going we the oxy sensor has to compare the exhaust gas with the reference air.
- Electrodes which are used here are platinum electrodes.
- And the electrolyte, as I mentioned just now, it's a solid state.
- Electrochemical fuel cell.
- Reference air and of course these electrodes which are platinum.
- two to point eight.
- What happens is we have oxygen at the cathode.
- That is the reason why we chose zirconium oxide doped with Y2O.
- Three because we know that when zirconium oxide is duped with Y2O three, it creates oxide ion vacancies in the electrolyte.
- And since we have oxide ion which is formed at the ca at cathode and this oxide ion has to jump through the electrolyte and reach the anode so we have to provide this electrolyte which is a solid electrolyte zirconium oxide doped with y2o3.
- So these are the reactions which happen at the anode and cathode, and we can see that both the reactions are similar because it is like a concentration cell.
- Both sides at the anode as well as the cathode we have the oxygen which is present at the cathode we have reduction happening giving you O2 minus these O2 minus ions are moving towards the anode.

Example or application from the lecture

- Let us study the construction of the oxygen sensor.
- Let us study the working of this oxygen sensor.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 15:00–19:59

And there they undergo oxidation, giving you O2 plus 4 electrolyte electrons.

Key details

- And the electrolyte is zirconium oxide doped with Y2O3.
- Solid electrolyte here because we need an electrolyte which is going to facilitate the movement of oxide ions.
- What would be the cell voltage?
- Now, since it is a concentration cell, we know that there will be no E naught, E naught value cancels values cancel for the anode and the cathode reactions, and therefore we have two.
- point three zero three r t over four f four is because of the number of electrons which are getting exchanged in these reactions so log of p1 by p2 now p1 is the partial pressure of oxygen in the reference air that is in the atmosphere and P2 is the partial pressure of oxygen in the exhaust.
- The reason for that is lead is going to poison the platinum catalyst.
- We know that we have used platinum at the anode and cathode.
- Poison the platinum catalyst, and that is going to reduce the efficiency of the oxygen sensor.
- When tetraethylaid is used because if the automobile is equipped with an oxygen sensor, this lead is going to poison the platinum and this will reduce the efficiency of the oxygen sensor.
- And we know that oxygen sensor is playing a very important role and therefore.
- Nowadays, unleaded gasoline is being used because most of the automobiles are equipped with oxygen sensor.
- For an oxygen sensor would be increased tailpipe emissions and increased fuel consumption and hesitation on acceleration.
- So in the next class we are going to talk about another set of energy storage devices and that is supercapacitors.

Example or application from the lecture

- The cell voltage can be given by the Nernst equation.
- Like for example methyl tertiary butyl ether.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

12 Source slide deck: Li ion battery LFP batteries

Study focus: Source slide deck: Li ion battery LFP batteries. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000294 · EV0000295

SHORT-NOTE RECAP

Study notes · Source slide deck: Li ion battery LFP batteries

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

- Department of Science and Humanities

Slide 2 · ENGINEERING CHEMISTRY

- Lithium-ion battery –(Lithium iron phosphate as cathode)

Slide 3 · ENGINEERING CHEMISTRY

- Lithium-ion battery (Lithium iron phosphate as cathode)
- Lithium ion moves from anode to cathode while discharging and from cathode to anode while charging
- Materials used as anode and cathode should be capable of lodging Lithium ions

Slide 4 · ENGINEERING CHEMISTRY

- Cathode (positive electrode) | Lithium iron phosphate, $LiFePO_4$ | Source and host of lithium ions
- Electrolyte | Lithium salt such as $LiPF_6$ dissolved in organic carbonate solvents | Conducts Li^+ ions between electrodes
- Separator | Microporous polymer, commonly polyethylene (PE) or polypropylene (PP) | Prevents direct electrical contact while allowing Li^+ transport

Slide 5 · ENGINEERING CHEMISTRY

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 6 · ENGINEERING CHEMISTRY

- At Anode: Li(1-x)FePO4 + xLi++ xe- →LiFePO4
- Net Reaction: xLiC6 + Li(1-x)FePO4 → LiFePO4 + xC6
- At Anode: LiFePO4 → Li(1-x)FePO4 + xLi++ xe

Slide 7 · ENGINEERING CHEMISTRY

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 8 · ENGINEERING CHEMISTRY

- Lighter than other rechargeable batteries for a given capacity
- Li-ion chemistry delivers a high open-circuit voltage of approximately 4 V and LFP battery delivers a voltage of about 3.2 V.
- Safety concerns if overheated or overcharged

Slide 9 · ENGINEERING CHEMISTRY

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 10 · Department of Science and Humanities

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000294 · EV0000295 · EV0000296 · +27 source units in pack.json

13 Source slide deck: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys

Study focus: Source slide deck: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000450 · EV0000451

SHORT-NOTE RECAP

Study notes · Source slide deck: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY

- Department of Science andA,B) Humanities

Slide 2 · ENGINEERING CHEMISTRY

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING CHEMISTRY

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 4 · ENGINEERING CHEMISTRY

- Abundant availability in •High cost
- Compatibility of hydrogen •It is still dependent on fossil

Slide 5 · ENGINEERING CHEMISTRY

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 6 · ENGINEERING CHEMISTRY

- Natural gas and steam is used to produce Hydrogen
- reaction with steam to produce additional H_2 and CO_2. Schematic
- representation of • The resulting gas mixture is then purified to obtain pure

Slide 7 · ENGINEERING CHEMISTRY

- Processes use light energy to split water into hydrogen
- uses electricity to split water into hydrogen and oxygen
- Method is promising option for carbon-free hydrogen

Slide 8 · ENGINEERING CHEMISTRY

- Green Hydrogen refers to production of Hydrogen gas
- from renewable resources, through photocatalysis
- Hydrogen production by electrolysis can be achieved by

Slide 9 · ENGINEERING CHEMISTRY

- Anodic materials: Ni and its alloys
- Electrolyte : KOH, NaOH (20-40 wt %)
- Operation temperatures are between 343 – 363 K

Slide 10 · ENGINEERING CHEMISTRY

- Pure water is the very poor conductor of electricity
- because splitting of water is very less, hence acid
- (H2SO4 in acid electrolyzer) or base (KOH or NaOH in

Slide 11 · ENGINEERING CHEMISTRY

- not happen on its own. It requires a minimum of
- $\Delta G^\circ = +237$ kJ/mol of Gibbs free energy, which
- translates to a minimum potential of 1.23 V per 11

Slide 12 · ENGINEERING CHEMISTRY

- Charge generation: Electrons get excited from VB to
- Surface redox: H^+ reduced to H2; water oxidised to O2.
- Oxidation in valance band : 2 H2O + 4 h^+ → O2 + 4 https://www.tandfonline.com/doi/full/10 .1080/01614940.2024.2446476

Slide 13 · ENGINEERING CHEMISTRY

- Band gap of the photocatalyst
- Surface area of the catalyst
- Use of cocatalysts (Pt, Ni, MoS_2)

Slide 14 · Slide page 14

- no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection.

EV0000450 · EV0000451 · EV0000452 · +37 source units in pack.json

14 Source slide deck: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides

Study focus: Source slide deck: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator.

EV0000490 · EV0000491

SHORT-NOTE RECAP

Study notes · Source slide deck: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

- Department of Science and Humanities

Slide 2 · ENGINEERING CHEMISTRY

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING CHEMISTRY

- The Hydrogen storage is crucial for effective utilization of Hydrogen for its versatile applications.
- Hydrogen can be stored as
- Physical and material-based methods of storage of Hydrogen

Slide 4 · ENGINEERING CHEMISTRY

- Hydrogen Storage in Metal-Organic Frameworks: A Review - research journal (gyanvihar.org)

Slide 5 · ENGINEERING CHEMISTRY

- Storage of hydrogen in the form of gas usually requires high-pressure tanks (350–700 bar tank pressure)
- Storage of hydrogen as a liquid requires cryogenic temperatures because the boiling point of hydrogen is $-252.8^{\circ}C$ at an atmospheric pressure

- Hydrogen can also be stored on the surfaces of solids (adsorption) or within solids (absorption)

Slide 6 · ENGINEERING CHEMISTRY

- Depending on the type of hydrogen being stored in host materials, solid-state hydrogen storage can be classified into two categories:
- Absorption of hydrogen atoms into the material
- Adsorption of Hydrogen molecules onto the surface of adsorbent

Slide 7 · ENGINEERING CHEMISTRY

- Some of the most prominent materials for Hydrogen Storage are
- Liquid organic hydrogen carriers (LOHC)

Slide 8 · ENGINEERING CHEMISTRY

- Representing a novel class of porous materials
- Enables high density energy storage of clean fuel gas in MOF adsorbents
- Zn-MOF (MOF-5) remains an attractive hydrogen storage material

Slide 9 · ENGINEERING CHEMISTRY

- LOHC technology has shown great potential for efficient and stable hydrogen storage and transport
- This technology allows for safe and economical large-scale transoceanic transportation and long-cycle hydrogen storage
- 1-s2.0-S2588913323000248-ga1_lrg.jpg (1161×886) (els-cdn.com)

Slide 10 · ENGINEERING CHEMISTRY

- Contain a larger quantity of hydrogen than the same volume of liquid hydrogen
- Have advantage in the amount of hydrogen on a weight basis
- Certain interstitial hydrides are very suitable for hydrogen storage and transportation

Slide 11 · ENGINEERING CHEMISTRY

- Can be compressed into high pressure in gas form
- hydrogen can be released whenever required by thermal stimulation or any other technique
- Cooling and compression process consume energy (net loss about 30%)

Slide 12 · Department of Science and Humanities

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000490 · EV0000491 · EV0000492 · +33 source units in pack.json

15 Source slide deck: Experiential learning Based on Hydrogen storage

Study focus: Source slide deck: Experiential learning Based on Hydrogen storage. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000637 · EV0000638

SHORT-NOTE RECAP

Study notes · Source slide deck: Experiential learning Based on Hydrogen storage

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

- Department of Science and Humanities

Slide 2 · ENGINEERING CHEMISTRY

- Term paper (Review article) - Hydrogen storage methods

Slide 3 · Department of Science and Humanities

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000637 · EV0000638 · EV0000639 · +6 source units in pack.json

16 Historical question-source pages · unit mapping candidates

Study focus: Historical question-source pages · unit mapping candidates. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000646 · EV0000647

SHORT-NOTE RECAP

Study notes · Historical question-source pages · unit mapping candidates

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Archive page 50 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.5833; source text/OCR may be partial. OCR/sparse page: visual confirmation is required. Principle Features and Content e) 1. Prevent Waste The primary goal is to synthesize the target product with little to no by- products or waste. Waste treatment adds to costs. 2. Atom Economy

Maximize the incorporation of reactants into the final product. Aim for 100% atom economy using addition/rearrangement reactions. 3. Safer Synthesis Minimize or avoid hazardous substances in synthesis. Example: Use CO, instead of toxic phosgene in polyurethane synthesis. 4. Safe Products Products should be effective but with reduced toxicity. Minimize harmful side effects on users and the environment. 5. Design Benign Chemicals Use safer solvents/reaction conditions. Avoid harmful solvents like chloroform; prefer water, CO,, ionic liquids, or solvent-free systems. 6. Design for Energy Efficiency Reduce energy intensity by using catalysts (e.g., bio-catalysts) and modern methods like microwave or ultrasound irradiation. 7. Use of Renewable Feedstock Prefer renewable raw materials like biomass over non-renewable sources such as coal and petroleum. 8. Reduce Derivatives Avoid unnecessary derivatization steps (e.g., blocking/protecting groups) to minimize waste and enhance atom efficiency. 9. Catalysis Use catalysts to accelerate reactions, reduce energy use, and achieve high yields with recyclable, selective reactions. 10. Design for Degradation Products should degrade into harmless substances after use to prevent environmental accumulation (e.g., pesticides, polymers). 11. Pollution Prevention Employ real-time monitoring and control of processes to prevent hazardous substances. Recycle unreacted chemicals. 12. Accident Prevention Choose chemicals and processes to minimize risks of accidents like fires, explosions, and toxic releases. Equip facilities with safety measures.

Archive page 4 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (5.0 Marks) (4.0 Marks) (5.0 Marks) (4.0 Marks) 3.a. Calculate the capacity (Ah), energy density (Wh/Kg) and electricity storage density (Ah/Kg) of Zn-air battery if 3.6g of Zn is stored in the battery weighing 75g. Given data: Voltage available from the battery = 1.39V and gram molar mass of Zn = 65.38 --(5M) 3.b. For Lithium-ion battery, provide the following details: (4M) Anode: Cathode: Reaction during Charging: Any two revalant disvantages: 3.c. For H2-O2 alkaline fuel cell, provide the following details: (5M) Anode: Cathode: Any two relevant advantahes: Why fuels feed should be free form CO ? 3.d. i. Working of Glucose Biosensor involves "Enzyme Glucose oxidase(GOx) catalyze the oxidation of glucose by molecular oxygen producing glucolactone and hydrogen peroxide" briefly explain using the reactions involved (2M) ii. Calculate the efficiency of the H -O fuel cell, if its cell potential is 1.235V (Given the enthalpy of formation of water is -285.83kj/mole, n=2). (2M) 2 2 2

Archive page 3 from 11__1zfeA5C-J7.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 2.d. Give any two factors which affect conductance of an electrolyte. 3.a. (i) Why should a polymer electrolyte membrance fuel cell be operated in the temperature range of 60 - 90 C ? (2 marks) (ii) Why can't carbon containing fuels be used in an alkaline fuel cell? (2 marks) (iii) Why is it that expensive catalyst need not be used in a solid oxide fuel cell? (2 marks) 3.b. (i) Calculate the energy density in Wh/Kg and power density in W/Kg for a battery which weighs 3 Kg. The battery has 120g of Zn as anode and discharges a constant current for 1800 minutes. The voltage output of the battery is 1.4V, the number of electrons involved in the redox reaction is 2 and atomic mass of ziinc is 65 and Faraday's constant is 96500 C/mol. (4 marks) (ii) Calculate the efficiency of H -O alkaline fuel cell, Given: E =1.20 V , no. of electrons involved in the redox reaction is 2 and enthalpy change accompanying the reaction ΔH = -285.8 kJ/mole. (2marks) 3.c. (i) How does an amperometric glucose biosensor work and give us the the concentration of glucose in the blood? Explain with relevant reactions. (No diagram required). 4 marks (ii) Give two points of difference between a battery and a supercapacitor. 2 marks 0 2 2 oCell

Archive page 4 from 1__1rBrYYqw3s.pdf; unit match candidate score 1.3333; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.b. i) With the help of neat labeled diagram explain the construction and working of H -O alkaline fuel cell. ii) Calculate the potential of the H -O alkaline fuel cell if its efficiency is 84.25 %. (Given the enthalpy of formation of water is -285.83 kJ/mole, F=96,500 c/mole.) 3.c. i) Mark the regions in which the following energy storage devices appear in the Ragone plot. a) Fuel cell (b) Li-ion battery (c) Supercapacitor ii) Mention any two materials used for hydrogen storage. 3.d. i) Discuss the working principle of glucose biosensor with relevant reactions. ii) Mention any one limitation of glucose sensor. 3.e. 50 cm of given hard water sample requires 15.6 cm of 0.075 M EDTA for complete reaction in the estimation of hardness by complexometry. Calculate the total hardness of water. 2 2 2 2 3 3

Archive page 3 from 2__11pj9jsL0V.pdf; unit match candidate score 1.25; source text/OCR may be partial. (6.0 Marks) (5.0 Marks) (2.0 Marks) i) Name the components of the H -O alkaline fuel cell. Calculate the efficiency of fuel the cell if ΔH = -285.5 kJ/mol and E = 1.21 V. ii) Draw a Ragone plot and indicate the position of the following energy conversion/storage devices: fuel cells, batteries, and supercapacitors. (7.0 Marks) 2.c. What is the galvanization? Mention the steps involved in the galvanization process and its advantages. 2.d. For the electrochemical phenomenon of corrosion. i). Write two reactions at the cathode surface in the presence of O ii). Write the formation of corrosion product if iron is exposed to ambient conditions. (Give reactions) Justify the following, i) Sodium meta arsenite is a cathodic inhibitor. 2.e. What are the factors affecting the conductance of an electrolyte? 3.a. 2. 2 2 formation (H2O) ocell

Archive page 4 from 2__11pj9jsL0V.pdf; unit match candidate score 1.0833; source text/OCR may be partial. (5.0 Marks) (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.b. For a Zinc-air battery : i) Draw a neat labeled diagram ii) Write the reactions taking place at the anode and cathode ii) Why do zinc-air batteries have high energy density? 3.c. What is the sensor? Mention the FIVE types of sensors with suitable examples. 3.d. Fill in the following table for the alkaline electrolysis process. (Hydrogen storage and production) Charge carrier: Anodic materials:
Cathodic materials: Electrolyte materials Catalyst materials: 3.e. What is intrinsic viscosity ? How is it related to the molecular mass of a polymer?

Archive page 4 from 3__1MwY1CTl_j.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 4/7 (3.0 Marks) (7.0 Marks) (7.0 Marks) 2.e. Discuss, with the help of Henderson-Hasselbalch equation, under which situation pH and pKa become equal 3.a. (a) Discuss the construction and working principles of Zn-air battery with a labelled diagram and the relevant chemical reactions (5M) (b) Calculate the electricity storage density (Ah/kg) of a Lithium battery which has 2.0 g of lithium as anodic material. The total weight of the battery is 65 g. (Given : Atomic mass of lithium is 7) (2M) 3.b. Discuss the construction and working principles of a H -O alkaline fuel cell with a clearly labelled diagram and the chemical reactions. Mention two disadvantages of H -O alkaline fuel cell (5M + 2M) 2 2 2 2

Archive page 5 from 3__1MwY1CTl_j.pdf; unit match candidate score 0.9167; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 5/7 (4.0 Marks) (4.0 Marks) (3.0 Marks) 3.c. (a) If the E for H -O alkaline fuel cell is 1.23 V and efficiency is 83%, calculate the heat evolved (in kilo Joules) during the reaction (2M)

(b) Why do supercapacitors have an enormous capacitance? (2M) 3.d. Discuss the construction and working principles of the oxygen sensor placed in the automobile exhaust stream with labelled diagram and chemical reactions 3.e. (a) Define viscosity coefficient of a liquid (1M) (b) Write Poiseuille's equation and explain all the terms (2M) oCell 2 2

Archive page 4 from 4__1eKMFaJ2lx.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 4/7 (2.0 Marks) (6.0 Marks) (6.0 Marks) 2.e. (i) Mention the role of sulphuric acid in the estimation of ferrous iron by potentiometric titration method. ii) Why pH is equals to pKa at half equivalence point in the pH metric titration of weak acid vs strong base ? 3.a. i) what are reserve batteries ? Give an example. ii) Calculate the capacity (Ah), energy density (Wh/kg) and power density (W/kg) of a battery if 2.4 g of lithium is stored in the battery and the battery lasts for 120 mins. The battery weighs 115 g and gives steady voltage of 3.4 V. (Given: F = 96500 C/mol, atomic mass of lithium = 7) 3.b. i) With the help of neat labeled diagram explain the construction and working of H -O alkaline fuel cell. ii) Calculate the potential of the H -O alkaline fuel cell if its efficiency is 84.25 %. (Given the enthalpy of formation of water is -285.83 kJ/mole, F=96,500 c/mole.) 2 2 2 2

Archive page 5 from 4__1eKMFaJ2lx.pdf; unit match candidate score 1.0833; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 5/7 (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.c. i) Mark the regions in which the following energy storage devices appear in the Ragone plot. a) Fuel cell (b) Li-ion battery (c) Supercapacitor ii) Mention any two materials used for hydrogen storage. 3.d. i) Discuss the working principle of glucose biosensor with relevant reactions. ii) Mention any one limitation of glucose sensor. 3.e. 50 cm of given hard water sample requires 15.6 cm of 0.075 M EDTA for complete reaction in the estimation of hardness by complexometry. Calculate the total hardness of water. 3 3

Archive page 4 from 5__1pIeTzocnU.pdf; unit match candidate score 0.9167; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 4/7 (2.0 Marks) (6.0 Marks) (6.0 Marks) 2.e. What are the two factors on which the absorbance of light of a particular wavelength by a substance in solution varies with? 3.a. i) Give any two reasons why secondary batteries have limited cycle life. ii) A battery using 1.7 g of lithium as anodic material gives a steady voltage of 3.6 V and lasts for 2 hours. If the weight of the battery is 50 g, then calculate the power density and energy density of the battery. [Given: Atomic mass of lithium is 7, F= 96500 C/mol] 3.b. i) If the E Cell for H -O alkaline fuel cell is 1.23 V the heat evolved during the reaction ΔH = -285.8 kJ/mole, calculate the efficiency of the fuel cell. (ii) Why can't carbon-containing fuels be used in an alkaline fuel cell? (iii) Porous carbon materials are used as electrode materials in supercapacitors. Give reason.

Archive page 5 from 5__1pIeTzocnU.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 5/7 (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.c. i) What is a biosensor? Mention the components required for the construction of glucose sensors. ii) Give any two points of difference between a potentiometric and amperometric sensor. 3.d. Mention the five different types of materials used for hydrogen storage. 3.e. Give any two advantages of the bottom-up solution combustion synthesis method for the synthesis of nanomaterials.

Archive page 4 from 6__1H6QH_PVNf.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 4/7 (2.0 Marks) (6.0 Marks) (6.0 Marks) 2.e. (i) Mention the role of sulphuric acid in the estimation of ferrous iron by potentiometric titration method. ii) Why pH is equals to pKa at half equivalence point in the pH metric titration of weak acid vs strong base ? 3.a. i) what are reserve batteries ? Give an example. ii) Calculate the capacity (Ah), energy density (Wh/kg) and power density (W/kg) of a battery if 2.4 g of lithium is stored in the battery and the battery lasts for 120 mins. The battery weighs 115 g and gives steady voltage of 3.4 V. (Given: F = 96500 C/mol, atomic mass of lithium = 7) 3.b. i) With the help of neat labeled diagram explain the construction and working of H -O alkaline fuel cell. ii) Calculate the potential of the H -O alkaline fuel cell if its efficiency is 84.25 %. (Given the enthalpy of formation of water is -285.83 kJ/mole, F=96,500 c/mole.) 2 2 2 2

Archive page 5 from 6__1H6QH_PVNf.pdf; unit match candidate score 1.0833; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 5/7 (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.c. i) Mark the regions in which the following energy storage devices appear in the Ragone plot. a) Fuel cell (b) Li-ion battery (c) Supercapacitor ii) Mention any two materials used for hydrogen storage. 3.d. i) Discuss the working principle of glucose biosensor with relevant reactions. ii) Mention any one limitation of glucose sensor. 3.e. 50 cm of given hard water sample requires 15.6 cm of 0.075 M EDTA for complete reaction in the estimation of hardness by complexometry. Calculate the total hardness of water. 3 3

Archive page 3 from 7__16pEjcSTVO.pdf; unit match candidate score 0.75; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 2.d. Give two factors that affect the conductance of an electrolyte. 3.a. For a zinc - air battery- (i) Give its anodic, cathodic and net cell reaction (ii) Give the reason for its high energy density (iii) What would happen if carbondioxide enters the cell (iv) Give its one important application 3.b. (i) Calculate the efficiency of H -O alkaline fuel cell Given: E =1.20 V, Enthaphy of formation of water ΔH = -285.8 kJ/mole (ii) Why is the capacitance of a supercapacitor much higher than that of a normal capacitor? 3.c. (i) How does an oxygen sensor respond to a lean mixture? Why should the stiochiometric air to fuel ratio be maintained at 14.7:1? (ii) Calculate the capacity of a lithium battery which stores 2.0 g of lithium, Given : Atomic mass of lithium is 7 and number of electrons involved in the redox reaction is 1 and Faraday's constant is 96500C/mol. 2 2 Cell

Archive page 4 from 8__1015pA2nzY.pdf; unit match candidate score 1.0833; source text/OCR may be partial. 3/25/25, 4:03 PM ESA - Dec 2024 - UE22CY151B/UE22CY151A (set- 1) about:blank 4/7 (4.0 Marks) (5.0 Marks) (4.0 Marks) 3.b. What are supercapacitors? Write one advantage, disadvantage and application of supercapacitors 3.c. i) Discuss the working principle of glucose biosensor with relevent reactions. ii) Mention any one limitation of glucose sensor. 3.d. i) Mark the regions in which the following energy storage devices appear in the Ragone plot. a) Fuel cell (b)Li-ion battery (c) Supercapacitor

Archive page 4 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 1.0833; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 4/6 (6.0 Marks) (6.0 Marks) (7.0 Marks) 3.b. (i) Give the Nernst equation corresponding to the output voltage generated in an 'Oxygen Sensor' and explain the terms. What

are the inferences one can draw from the output voltage values and how is it useful. (4 marks) (ii) Why is Polymer electrolyte membrane fuel cell operated in the temperature range of 60 -90 C. (2 marks) 3.c. (i) Give the synthesis of hydrogen using the steam reforming process with methane as the starting material. (Mention the conditions of temperature, pressure and catalyst used in the reaction). (4 marks) (ii) How are 'Liquid organic hydrogen carriers (LOHC)' used in storage and transport of hydrogen gas. Give one example of LOHC. (2 marks) 3.d. (i) Of all the energy storage / conversion devices discussed, give a suitable one for the following applications- (4 marks) Hearing aids Mobiles and laptops Space exploration In hybrid cars for start-stop application (ii) Calculate the viscosity average molecular weight of polystyrene (atactic) from the following data: Intrinsic Viscosity =0.73, K = 11.5x10−5 and Mark Houwink parameter (a) is 0.73. (3 marks) 0

Archive page 4 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 4/7 (3.0 Marks) (7.0 Marks) (6.0 Marks) 2.e. i) Why colorimetric estimation of copper is done at 620 nm? (1M) ii) Draw the nature of the graphs in the determination of pKa value of weak acid. (2M) 3.a. For a Zn-air battery i) What are the materials used as anode, cathode, electrolyte and separator? (2M) ii) Write the reactions taking place at anode and cathode when the battery is discharging. (2M) iii) Define energy density of a battery and calculate energy density of zinc- air battery in Wh/kg. If 6.6g of Zn is stored in the battery weighing 36.8g. (Given; voltage of the given Zinc-air battery is 1.3 V and gram molar mass of zinc is 65.38) (3M) 3.b. i) Explain the construction and working of H -O alkaline fuel cell. (4M) ii) Calculate the efficiency of the H -O alkaline fuel cell. (Given; the enthalpy of formation of water is -285.83 kj/mole, E =1.19V). (2M) 2 2 2 2 cell

Archive page 5 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 5/7 (4.0 Marks) (6.0 Marks) (2.0 Marks) 3.c. i) Mention the advantages and Limitations of Steam reforming method of hydrogen production. (2M) ii) "Liquid organic hydrogen carriers (LOHC) can be used as storage media for hydrogen" Justify the statement with example. (2M) 3.d. What is an oxygen sensor? Give the construction, working of oxygen sensor and explain its role in automobiles. 3.e. Calculate the viscosity average molecular weight of a polymer solution by the following data. K=45.3x10 , a=0.64, intrinsic viscosity=0.75 -5

EV0000646 · EV0000647 · EV0000648 · +16 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__46f678f584cf1e30.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__46f678f584cf1e30.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__46f678f584cf1e30.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__46f678f584cf1e30.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Numericals_Energy storage devices transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_1__b31208265c8084c7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_1__b31208265c8084c7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_1__b31208265c8084c7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_1__b31208265c8084c7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9972; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9968; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Supercapacitors , Ragone plot transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9973; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Batteries Components and working of a battery classification of batteries primary secondary and reserve batter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Batteries Components and working of a battery classification of batteries primary secondary and reserve batter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Batteries Components and working of a battery classification of batteries primary secondary and reserve batter (converted PDF; original hash d9c42f3a686ef40d0ad83248232738371df224c923330cc30c846a0fe086e2ba)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Battery characteristics Numericals based on battery characteristics	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Battery characteristics Numericals based on battery characteristics	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Battery characteristics Numericals based on battery characteristics (converted PDF; original hash d969bdbbc833f7c51be9da5bbb4bfe7b42a2df103a0bab422c5e056b6f825668)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Modern batteries Zn air battery Li metal batteries	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Modern batteries Zn air battery Li metal batteries	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Modern batteries Zn air battery Li metal batteries (converted PDF; original hash bdc805e3e05fab8565464d114d48cf4c1b7038bda5f4a1e5861ff359c1b7d021)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Li ion battery LFP batteries	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Li ion battery LFP batteries	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Li ion battery LFP batteries (converted PDF; original hash e7d3c3ad6b1affd5f4b6b070ba5abc52a531d271f5ffaa87ba3cc92ac0bc6425)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Fuel cells Principle and working H2 O2 alkaline fuel cell	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Fuel cells Principle and working H2 O2 alkaline fuel cell	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Fuel cells Principle and working H2 O2 alkaline fuel cell (converted PDF; original hash bf5f43fd3bc7b37ccd89e3b36ad6afdb975dd6a975cc6895f1df8efd31603657)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel (converted PDF; original hash 6248c89e8ebd35030a04cc8e9da52a8570ca4eca0fa27c7945a1729440651644)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Supercapacitors construction and working Ragone plot	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Supercapacitors construction and working Ragone plot	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Supercapacitors construction and working Ragone plot (converted PDF; original hash 91f6dd1e8f435dea9f720bbbc2eb56a2bfa46fd1a26c6dd5229b080cf303d6d3)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys (converted PDF; original hash bab821775eabfe2b59e29b4273d5a699b07265c9a08ef00d7ffe97ec6df36ef3)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides (converted PDF; original hash 52d83559f0c2af524e16a30cc9ba96ba11682dcdd8566e52ead1c28dfa273bc6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode (converted PDF; original hash 91c0f334bd8f387fe672ecac12ac8855df52bc5c66dcb7560d2ce9d78698005e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor (converted PDF; original hash 7f4ee6571c560e964fab20c01797fd6cf48c0ce32ee23b3a61dce064c5c46476)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Based on Hydrogen storage	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Experiential learning Based on Hydrogen storage	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Based on Hydrogen storage (converted PDF; original hash 67f035536e4cb2e700cb78ee432824ffb622d60f3e7a2a975fd8d831efdbcf55)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: Chem_Extra_Topics__1-HfOeZNwo.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Extra/Chem_Extra_Topics__1-HfOeZNwo.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: 10__1vFS6E1bKV.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/10__1vFS6E1bKV.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 11__1zfeA5C-J7.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/11__1zfeA5C-J7.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 1__1rBrYYqw3s.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/1__1rBrYYqw3s.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 2__11pj9jsL0V.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/2__11pj9jsL0V.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 3__1MwY1CTl_j.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/3__1MwY1CTl_j.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 4__1eKMFaJ2lx.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/4__1eKMFaJ2lx.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 5__1pIeTzocnU.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/5__1pIeTzocnU.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 6__1H6QH_PVNf.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/6__1H6QH_PVNf.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 7__16pEjcSTVO.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/7__16pEjcSTVO.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: 8__1015pA2nzY.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/8__1015pA2nzY.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-EC[2025]__1ZtWpoG9Dv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-EC[2025]__1ZtWpoG9Dv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-RR[2025]__1xN30tqzNj.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-RR[2025]__1xN30tqzNj.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.

Source	Unit counts	Example decisions
SRC00002	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00003	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00005	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00006	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00007	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00009	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00014	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare...
SRC00015	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare...
SRC00017	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare...
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare...
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00034	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00035	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00037	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00038	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00039	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00040	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00041	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00042	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00044	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00043	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00045	0 covered · 0 excluded · 14 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00047	0 covered · 0 excluded · 14 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00046	0 covered · 0 excluded · 14 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00048	0 covered · 0 excluded · 12 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00050	0 covered · 0 excluded · 12 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00049	0 covered · 0 excluded · 12 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00051	10 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00053	10 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00052	10 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00054	0 covered · 0 excluded · 12 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00056	0 covered · 0 excluded · 12 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00055	0 covered · 0 excluded · 12 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00057	0 covered · 0 excluded · 14 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00059	0 covered · 0 excluded · 14 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00058	0 covered · 0 excluded · 14 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00060	0 covered · 0 excluded · 16 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00062	0 covered · 0 excluded · 16 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00061	0 covered · 0 excluded · 16 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00063	13 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00065	14 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00064	13 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00066	12 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00068	12 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00067	12 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00069	0 covered · 0 excluded · 20 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00071	0 covered · 0 excluded · 20 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00070	0 covered · 0 excluded · 20 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00072	0 covered · 0 excluded · 17 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00074	0 covered · 0 excluded · 17 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00073	0 covered · 0 excluded · 17 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00075	3 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00077	3 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00076	3 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00078	1 covered · 0 excluded · 0 gaps	q50: Original archive PDF page 50; candidate unit association is text-based and unconfirmed.
SRC00079	1 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
SRC00080	1 covered · 0 excluded · 0 gaps	q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00081	1 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
SRC00082	2 covered · 0 excluded · 0 gaps	q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed. q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
SRC00083	2 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed. q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
SRC00084	2 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed. q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.

Source	Unit counts	Example decisions
SRC00085	2 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed. q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
SRC00086	2 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed. q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
SRC00087	1 covered · 0 excluded · 0 gaps	q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00088	1 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
SRC00089	1 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
SRC00090	2 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed. q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.

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